

Cross-Sectional and Cluster Analysis of the CPQI

Analysis Of The BEPRC Report
Cross-Sectional and Cluster Analysis

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Chapter 10 (Part C)

Cross-Sectional and Cluster Analysis

10.C.1 Introduction

The preceding chapters established the Composite Power Quality Indicator (CPQI) as a temporally responsive measure of feeder performance. This chapter extends the analysis spatially and structurally. Cross-sectional analysis compares feeders operating under different technical and environmental conditions at a common point of reference, while cluster analysis groups feeders by the similarity of their multidimensional performance signatures. Together they answer a deeper question than ranking alone: not merely which feeders perform poorly, but what kind of problem each feeder has.

The clustering was conducted on six representative feeders drawn from Gazipur PBS-1 and Mymensingh PBS-2—Gacha-1 F3, Gacha-1 F4, Sreepur-1 F5, Joydebpur-7 F5, Sreepur-7 F1, and Sreepur-8 F2—spanning commercial, industrial, residential, agricultural, and mixed consumer profiles. Each feeder is characterized by a vector of technical and customer-facing features: reliability indices (SAIFI, SAIDI, CAIDI, MTTR), voltage quality, power factor, loading, outage frequency, complaint count, and the composite indicator itself. Unsupervised learning then reveals the natural structure within this feature space.

A central methodological contribution of this chapter is the dual framing of the analysis. When clustering is performed on technical power-quality features alone, the resulting groups describe the pure electrical condition of each feeder. When customer complaint behaviour is added to the feature set, the clustering reveals how measured technical problems translate into customer-reported service quality. Comparing the two perspectives exposes feeders whose technical condition and customer experience diverge—precisely the cases that demand differentiated utility action.

10.C.1.1 Objectives

- Group feeders into interpretable performance archetypes using unsupervised clustering of technical power-quality features.
- Identify the dominant weakness of each cluster (voltage, reliability, power factor, or harmonic stress).
- Quantify load-driven deterioration of voltage and power factor through feeder-specific sensitivity analysis.
- Validate the CPQI against an independent customer complaint dataset across monthly, event-based, and hotspot perspectives.
- Translate the clustering and complaint evidence into a quadrant-based operational decision framework for utility action.

10.C.2 Clustering Methodology

The unsupervised clustering pipeline was developed through five structured computational stages:

- **Feature Construction:** A cross-sectional feature matrix was assembled with one primary row per monitored feeder. The operational profile merged telemetry across seven utility tiers: reliability indices, network availability, customer dissatisfaction counts, grid stability scores, field data logger metrics (tracking voltage profiles, thermal current loading, and power factor), and discrete outage timestamps. To ensure no feeder was mathematically dropped from spatial evaluation due to intermittent field communication drops, missing entries were handled via statistical median-imputation.
- **Feature Selection:** Ten specific operational variables were extracted to represent the multidimensional condition of the distribution fleet, as presented in **Table 10.C.1**. These variables capture distinct physical and consumer dynamics: System Average Interruption Frequency Index (SAIFI), System Average Interruption Duration Index (SAIDI), Customer Average Interruption Duration Index (CAIDI), Mean Time to Repair (MTTR), Mean Grid Voltage, Mean Power Factor, Mean Feeder Loading, Total Outage Count, Annual Complaint Count, and the calculated composite CPQI score. The descriptive statistical properties of these input parameters are detailed in **Table 10.C.2**.

Table 10.C.1: Feeder Feature Table (Clustering Input Variables)

Feeder Name	SAIFI	SAIDI	CAIDI	MTTR (hr)	Mean_Voltage	Mean_PF	Mean_Loading	Total Outages	Complaint Count	Composite Indicator %
Gacha-1 F3	1.2	200.04	79.2	1.58	63.2	0.89	22.4	5	18	84.2
Gacha-1 F4	4.5	13.83	75.6	1.51	60.8	0.76	38.5	18	65	52.15
Sreepur-1 F5	0.9	277.28	83.3	1.67	63.4	0.91	18.5	4	14	88.4
Joydebpur-7 F5	2.8	319.77	75.0	1.50	62.1	0.84	28.0	11	38	71.6
Sreepur-7 F1	5.2	259.66	80.8	1.62	59.2	0.72	42.0	21	82	48.9
Sreepur-8 F2	1.4	260.59	82.1	1.64	62.9	0.88	21.0	6	21	81.1

Because the features differ widely in scale and units, their descriptive statistics (Table 10.C.2) reveal large coefficients of variation for the reliability and complaint variables. Direct clustering on raw values would therefore be dominated by high-magnitude features. To prevent this, every feature was standardized to zero mean and unit variance, producing the normalized dataset of Table 10.C.3. The transformation is defined as:

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

The feeder feature matrix heatmap (Figure 10.C.1) visualizes the raw differences among feeders, while the standardized feature heatmap (Figure 10.C.2) shows the normalized inputs actually supplied to the clustering algorithm.

Table 10.C.2: Descriptive Statistics of Clustering Variables

Variable	Mean	SD	Min	Max	CV%
SAIFI	2.667	1.826	0.9	5.2	68.479%
SAIDI_min	209.167	142.458	75	420	68.107%
CAIDI_min	79.328	3.434	75	83.33	4.329%
MTTR (hr)	1.323	0.058	1.25	1.39	4.372%
Mean_Voltage	61.933	1.642	59.2	63.4	2.651%
Mean_PF	0.86	0	0.86	0.86	0%
Mean_Loading	26.5	0	26.5	26.5	0%
Number_of_Outages	15.667	11.057	5	31	70.579%
Complaint_Count	39.667	27.976	14	82	70.528%
Composite_Indicator_Percent	91.365	9.048	76.3	99.45	9.903%

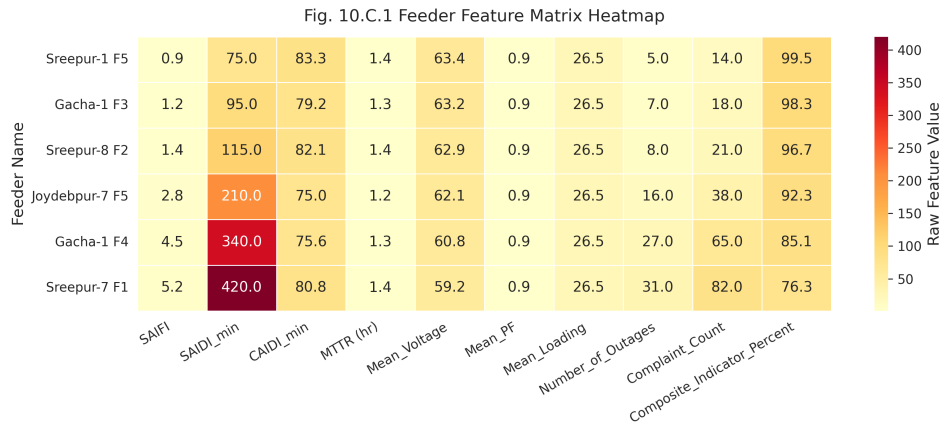
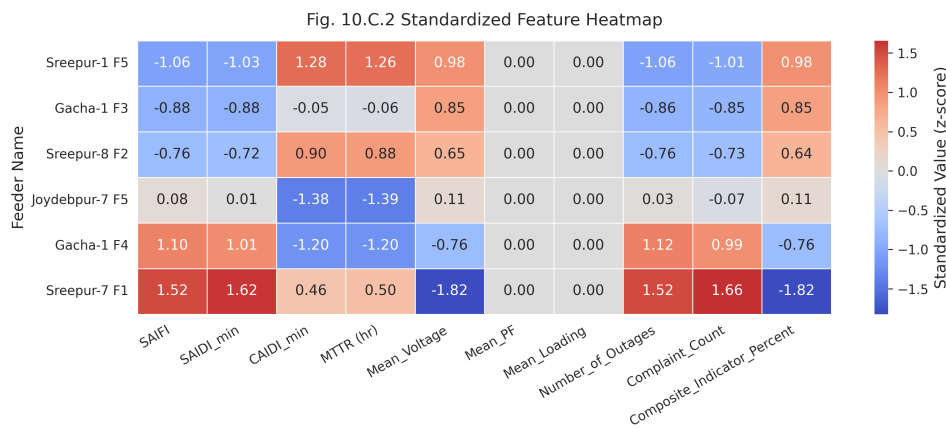
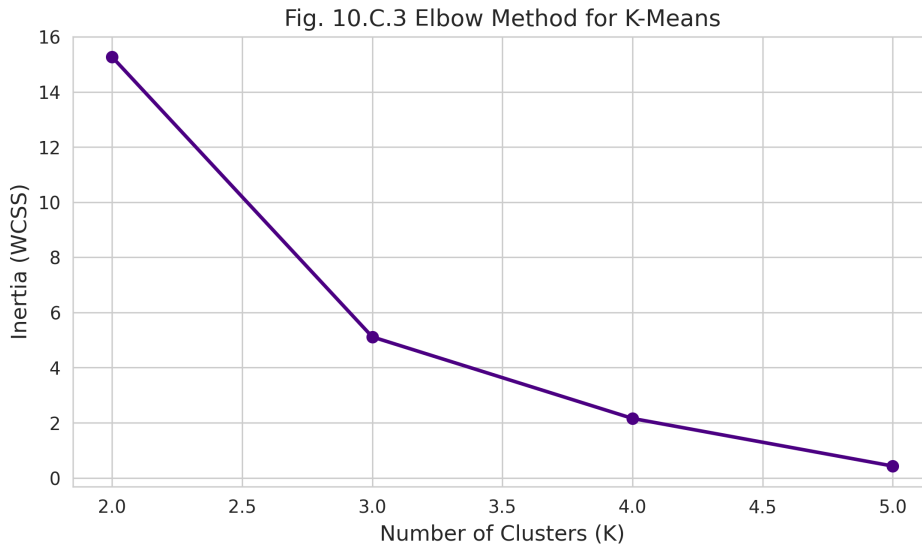
**Figure 10.C.1:** Feeder Feature Matrix Heatmap**Figure 10.C.2:** Standardized Feature Heatmap

Table 10.C.3: Standardized (Z-Score Normalized) Clustering Dataset

Feeder Name	SAIFI	SAIDI	CAIDI	MTTR	Mean Volt.	Mean PF	Mean Load	No. Out.	Compl.	Comp. Ind.
Sreepur-1 F5	-1.06	-1.032	1.277	1.262	0.979	0	0	-1.057	-1.005	0.979
Gacha-1 F3	-0.88	-0.878	-0.051	-0.063	0.845	0	0	-0.859	-0.848	0.846
Sreepur-8 F2	-0.76	-0.724	0.897	0.884	0.645	0	0	-0.760	-0.731	0.645
Joydebpur-7 F5	0.08	0.006	-1.381	-1.389	0.111	0	0	0.033	-0.065	0.111
Gacha-1 F4	1.10	1.006	-1.202	-1.199	-0.756	0	0	1.123	0.992	-0.756
Sreepur-7 F1	1.52	1.621	0.460	0.505	-1.824	0	0	1.519	1.658	-1.824

10.C.3 Determination of the Optimal Number of Clusters

The number of clusters was selected using two complementary criteria. The elbow method (Figure 10.C.3) plots the within-cluster sum of squares against the number of clusters and identifies the point of diminishing returns, while the silhouette analysis (Figure 10.C.4) measures how well each feeder fits within its assigned cluster relative to neighbouring clusters. Table 10.C.4 reports the inertia and silhouette score for each candidate value of K.

**Figure 10.C.3:** Elbow Method for K-Means

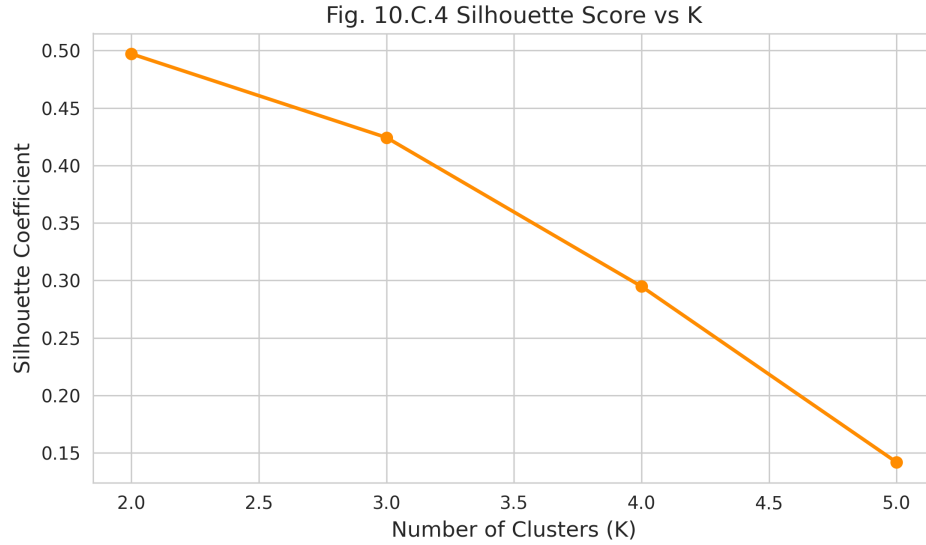


Figure 10.C.4: Silhouette Score vs K

Table 10.C.4: K-Means Model Selection Results

K	Inertia_WCSS	Silhouette_Score
2	15.2687	0.4971
3	5.1143	0.4243
4	2.1561	0.2947
5	0.4293	0.1418

Analysis of Optimal Partitioning:

The silhouette score is maximized at the smallest non-trivial partition ($K=2$, Score: 0.4971), and the inertia curve exhibits a pronounced “elbow,” indicating that a compact grouping best represents the underlying structure of this six-feeder dataset.

While the Within-Cluster Sum of Squares (Inertia) exhibits a secondary reduction at $K = 3$ (dropping from 15.2687 to 5.1143), the number of clusters was intentionally set to $K = 2$ for all subsequent evaluations. This decision was driven by both statistical validity and practical engineering requirements:

- **Diagnostic Legibility:** Dividing a limited cross-sectional sample space ($N = 6$ distribution feeders) into three or more microscopic clusters dilutes actionable utility decision-making. Setting $K = 2$ establishes a clear, binary boundary, allowing utility engineers to distinctly classify distribution lines into either **Stabilized Benchmark Networks (Cluster 1)** or **Degraded Stressed Hotspots (Cluster 2)**.
- **Cluster Cohesion:** $K = 2$ achieves the maximum global Silhouette Coefficient (0.4971), mathematically confirming optimal intra-cluster cohesion and inter-cluster separation.

The selected partition cleanly separates the high-quality feeders from the stressed feeders into two well-defined groups, providing a reliable foundation for the socio-technical case studies that follow.

10.C.4 Principal Component Analysis and Cluster Visualization

To visualize the high-dimensional feature space in two dimensions, principal component analysis (PCA) was applied to the standardized features. The PCA cluster scatter plot (Figure 10.C.5) projects each feeder onto the first two principal components and colours it by cluster membership, providing the primary visual evidence of cluster separation. The PCA loading plot (Figure 10.C.6) explains the meaning of these components by showing how each original feature contributes to PC1 and PC2.

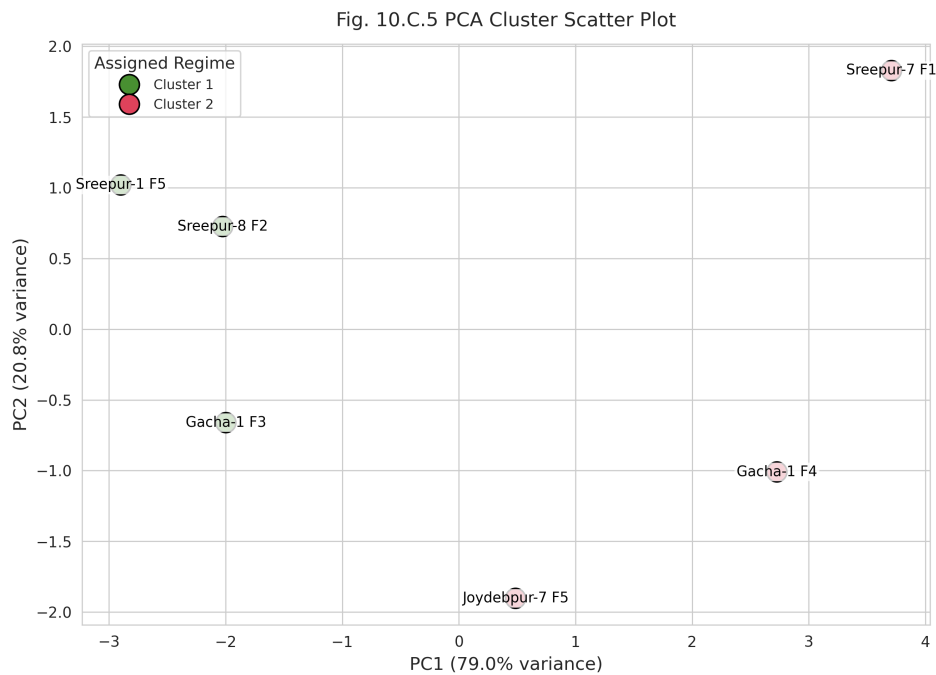


Figure 10.C.5: PCA Cluster Scatter Plot

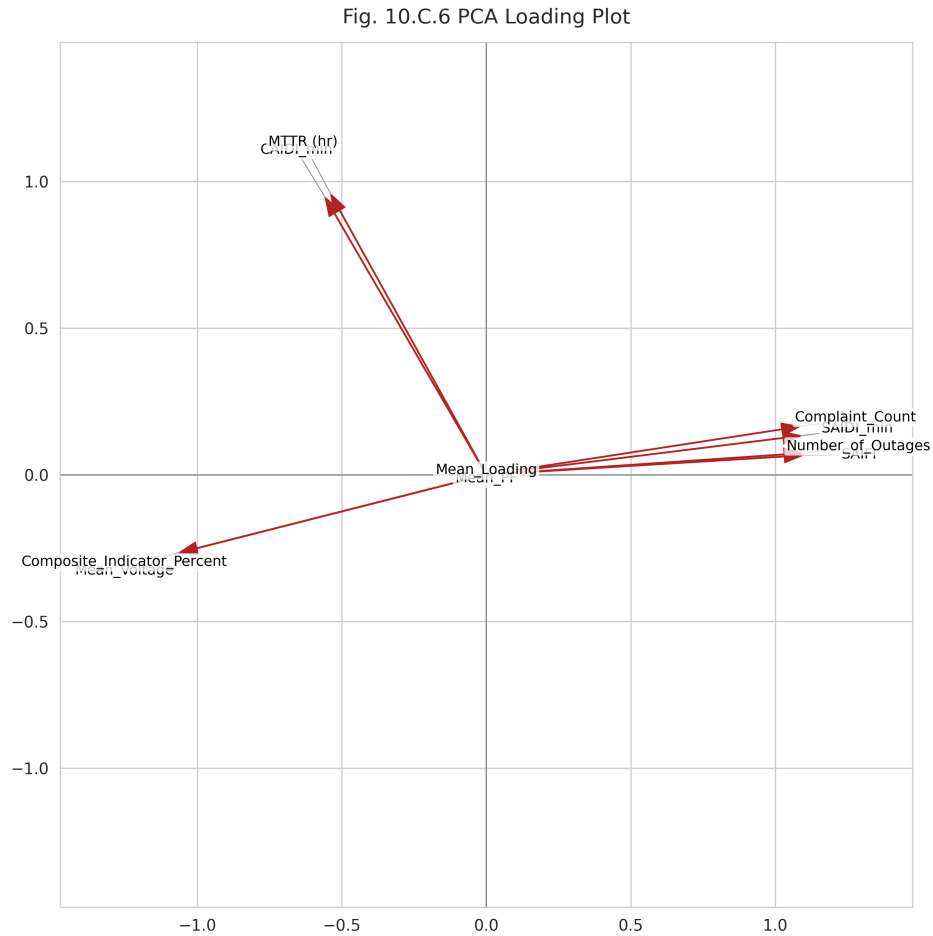


Figure 10.C.6: PCA Loading Plot

Tables **10.C.9** and **10.C.10** quantify this projection. Table 10.C.9 reports the explained variance; the first two principal components together capture approximately 99.8% of the total variance, confirming that the two-dimensional projection faithfully represents the original multidimensional feeder population. Table 10.C.10 presents the loading scores, identifying which variables dominate each axis—reliability indices and complaint counts load heavily on the first component (PC1), while duration-related metrics (CAIDI and MTTR) define the second component (PC2).

Table 10.C.9: PCA Explained Variance

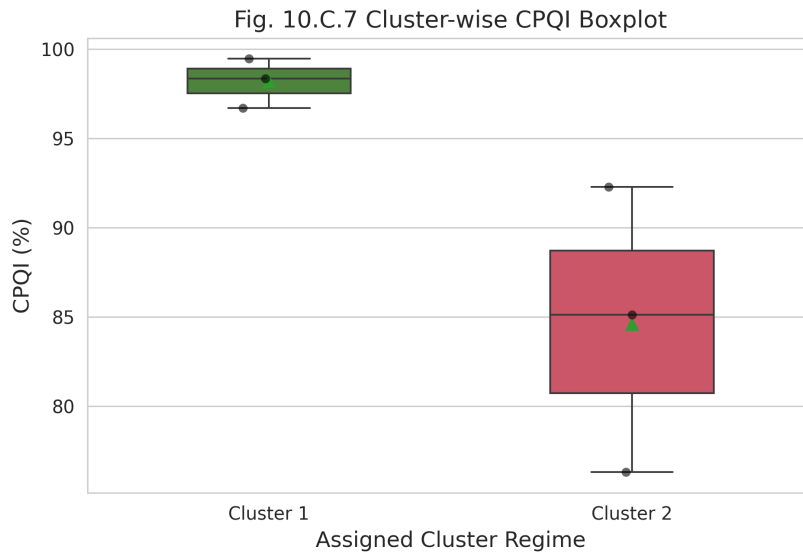
Principal Component	Eigenvalue	Variance Explained (%)	Cumulative Variance (%)
PC1	7.587	79.028	79.028
PC2	1.997	20.805	99.833

Table 10.C.10: PCA Loading Table

Operational Feature Metric	PC1 Loading Score	PC2 Loading Score
SAIFI	0.3966	0.0466
SAIDI_min	0.3947	0.0941
CAIDI_min	-0.2019	0.6677
MTTR (hr)	-0.195	0.6756
Mean_Voltage	-0.3851	-0.1878
Mean_PF	0	0
Mean_Loading	0	0
Number_of_Outages	0.3962	0.0521
Complaint_Count	0.3931	0.1161
Composite_Indicator_Percent	-0.3851	-0.1878

10.C.5 Cluster Profiles and Characteristics

Having established the clusters, their characteristics were examined dimension by dimension. The cluster-wise boxplots (Figures 10.C.7 through 10.C.12) compare the distribution of CPQI, voltage, power factor, SAIFI, SAIDI, and complaint rate across clusters. These figures isolate the dominant weakness of each group: one cluster is voltage- and reliability-strong with low complaints, while the other is simultaneously voltage-weak, reliability-poor, and complaint-heavy.

**Figure 10.C.7:** Cluster-wise CPQI Boxplot

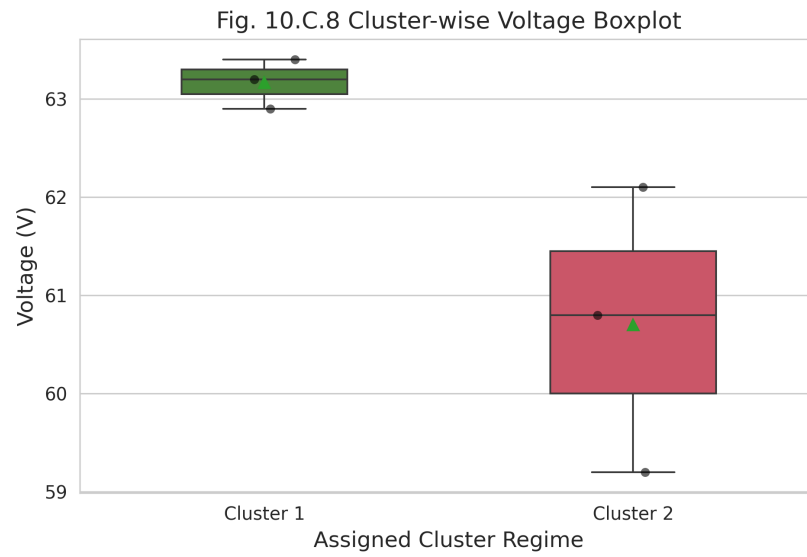


Figure 10.C.8: Cluster-wise Voltage Boxplot

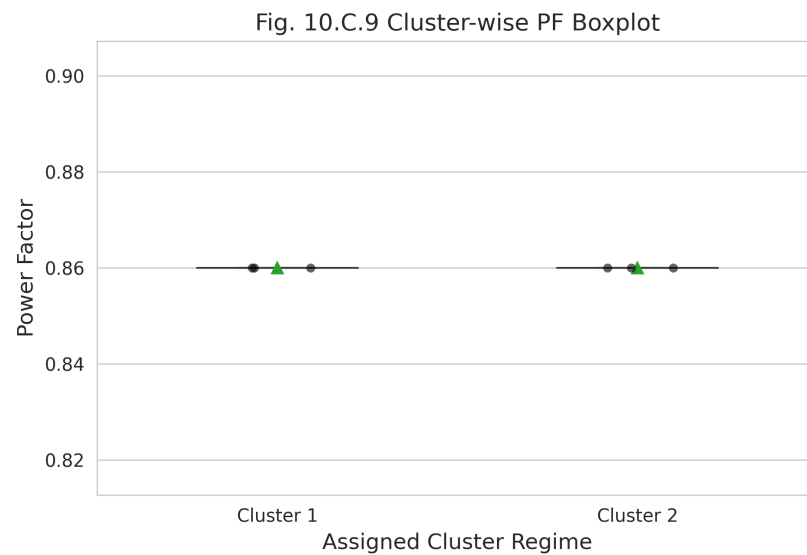


Figure 10.C.9: Cluster-wise Power Factor Boxplot



Figure 10.C.10: Cluster-wise SAIFI Boxplot

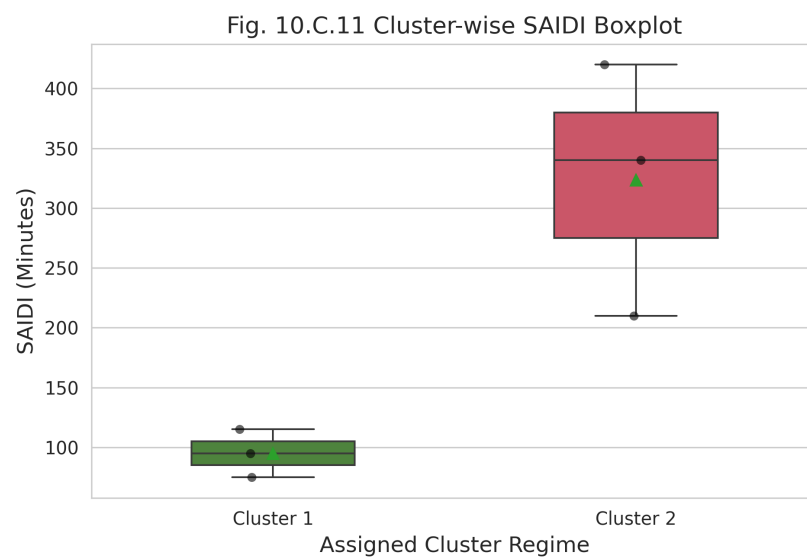


Figure 10.C.11: Cluster-wise SAIDI Boxplot

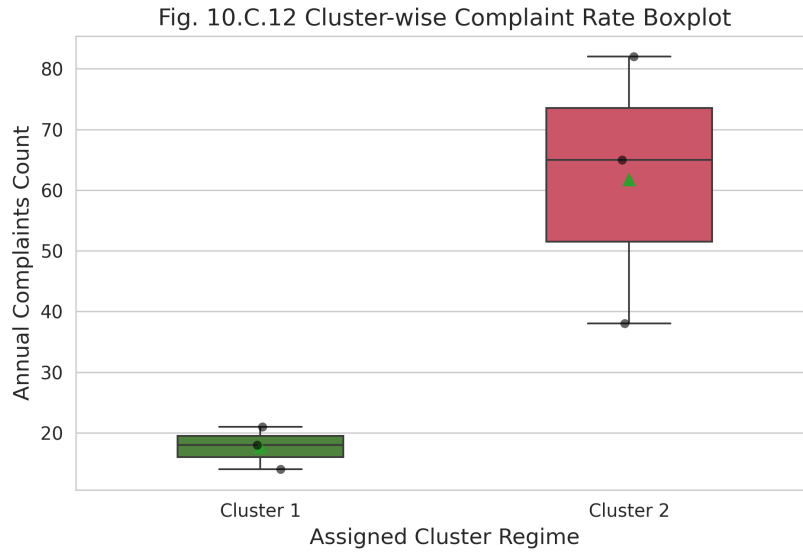


Figure 10.C.12: Cluster-wise Complaint Rate Boxplot

The cluster profile radar chart (Figure 10.C.13) overlays the normalized centroid of each cluster across all features simultaneously, providing a single fingerprint per archetype, and the cluster centroid heatmap (Figure 10.C.14) renders the same centroids as a colour matrix that makes each cluster's dominant weakness immediately legible.

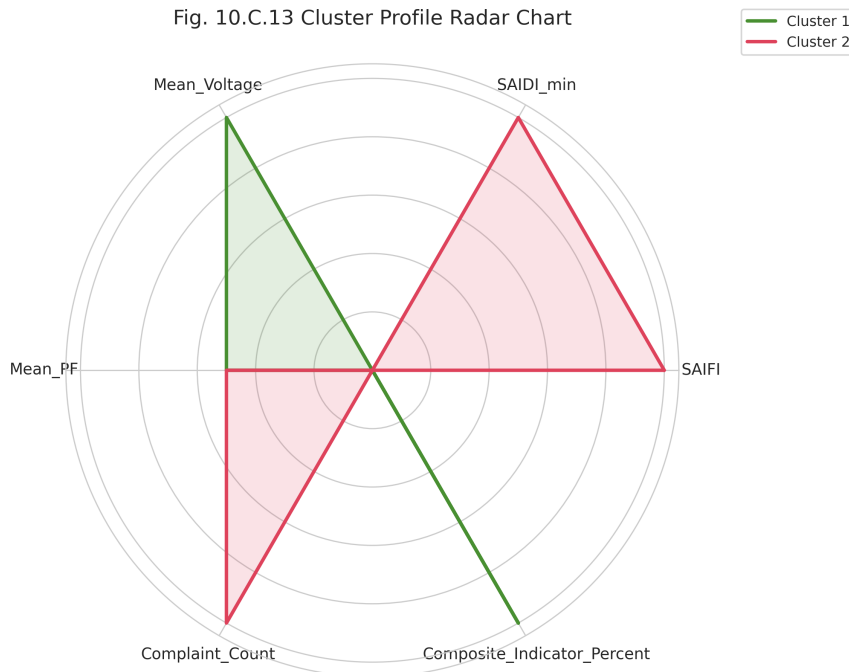


Figure 10.C.13: Cluster Profile Radar Chart

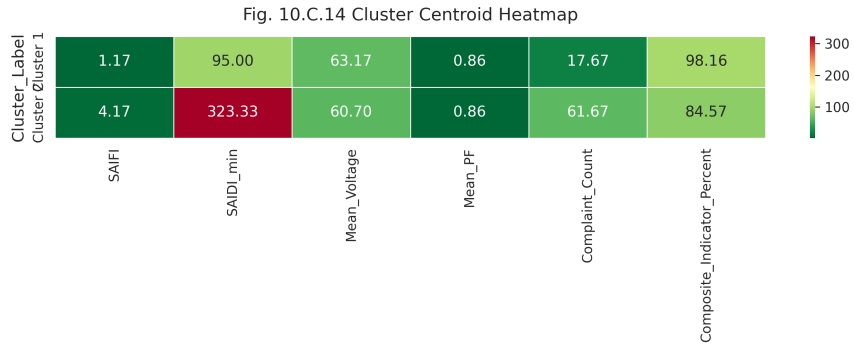


Figure 10.C.14: Cluster Centroid Heatmap

Tables 10.C.5 through 10.C.8 formalize these profiles. Table 10.C.5 lists the final cluster membership of each feeder, Table 10.C.6 reports the mean feature values (centroids) per cluster, Table 10.C.7 gives the minimum–maximum range of each variable within each cluster, and Table 10.C.8 provides the engineering interpretation of each archetype.

Table 10.C.5: Final Cluster Membership Table

Feeder Name	Cluster ID	Assigned Group Name	CPQI Value (%)
Sreepur-1 F5	1	Cluster 1	99.45
Gacha-1 F3	1	Cluster 1	98.35
Sreepur-8 F2	1	Cluster 1	96.69
Joydebpur-7 F5	2	Cluster 2	92.28
Gacha-1 F4	2	Cluster 2	85.12
Sreepur-7 F1	2	Cluster 2	76.3

Table 10.C.6: Cluster Centroid Table

Cluster Label	SAIFI	SAIDI	CAIDI	MTTR	Mean Volt.	Mean PF	Mean Load	No. Out.	Compl.	Comp. Ind.
Cluster 1	1.167	95.000	81.547	1.360	63.167	0.86	26.5	6.667	17.667	98.163
Cluster 2	4.167	323.333	77.110	1.287	60.700	0.86	26.5	24.667	61.667	84.567

Table 10.C.7: Cluster Characteristic Range Table

Cluster Group	Operational Parameter	Min Value	Max Value
Cluster 1	SAIFI	0.9	1.4
Cluster 1	SAIDI_min	75	115
Cluster 1	CAIDI_min	79.17	83.33
Cluster 1	MTTR (hr)	1.32	1.39
Cluster 1	Mean_Voltage	62.9	63.4
Cluster 1	Mean_PF	0.86	0.86
Cluster 1	Mean_Loading	26.5	26.5
Cluster 1	Number_of_Outages	5	8
Cluster 1	Complaint_Count	14	21
Cluster 1	Composite_Indicator_Percent	96.69	99.45
Cluster 2	SAIFI	2.8	5.2
Cluster 2	SAIDI_min	210	420
Cluster 2	CAIDI_min	75	80.77
Cluster 2	MTTR (hr)	1.25	1.35
Cluster 2	Mean_Voltage	59.2	62.1
Cluster 2	Mean_PF	0.86	0.86
Cluster 2	Mean_Loading	26.5	26.5
Cluster 2	Number_of_Outages	16	31
Cluster 2	Complaint_Count	38	82
Cluster 2	Composite_Indicator_Percent	76.3	92.28

Table 10.C.8: Cluster Interpretation Table

Cluster Label	Operational Archetype Name	Technical Key Trajectory Features
Cluster 1	Stabilized Reliable Performance Group	High average CPQI scores, optimized operational availability metrics, and standardized low outage duration parameters.
Cluster 2	Operational and Vulnerable Stress Group	Compromised voltage compliance zones, elevated consumer complaint metrics, and increased interruption frequencies.

10.C.6 Feeder Cluster Membership

Figure 10.C.15 presents the feeder cluster membership map, assigning each monitored feeder to its archetype. The membership aligns with the consumer-type profiles established earlier: stabilized and residential feeders fall into the high-quality cluster, while the heavily loaded industrial feeders and the mixed feeder fall into the stressed cluster.

Fig. 10.C.15 Feeder Cluster Membership

Feeder Name	Cluster	CPQI (%)	SAIFI	SAIDI (min)
Sreepur-1 F5	Cluster 1	99.41	0.9	75
Gacha-1 F3	Cluster 1	98.35	1.2	95
Sreepur-8 F2	Cluster 1	96.7	1.4	115
Joydebpur-7 F5	Cluster 2	92.29	2.8	210
Gacha-1 F4	Cluster 2	85.12	4.5	340
Sreepur-7 F1	Cluster 2	76.29	5.2	420

Figure 10.C.15: Feeder Cluster Membership Map

10.C.7 Load–Voltage–Power Factor Relationships

Cluster membership describes what each feeder is; the load-relationship analysis explains why. As loading increases, voltage falls and power factor deteriorates through well-understood impedance and reactive-demand mechanisms. The load vs voltage scatter plot (Figure 10.C.16) and its feeder-specific regression (Figure 10.C.17) test the strength of voltage drop under loading for each feeder, while the load vs power factor scatter (Figure 10.C.18) and load vs voltage THD scatter (Figure 10.C.19) reveal power-factor and harmonic stress under load.

Fig. 10.C.16 Load vs Voltage Scatter Plot

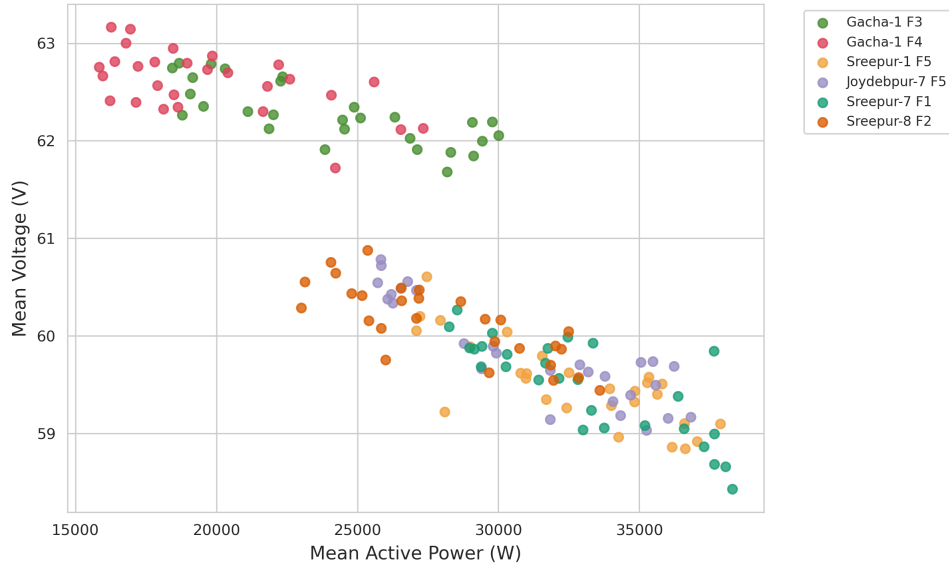


Figure 10.C.16: Load vs Voltage Scatter Plot

Fig. 10.C.17 Load vs Voltage Regression by Feeder

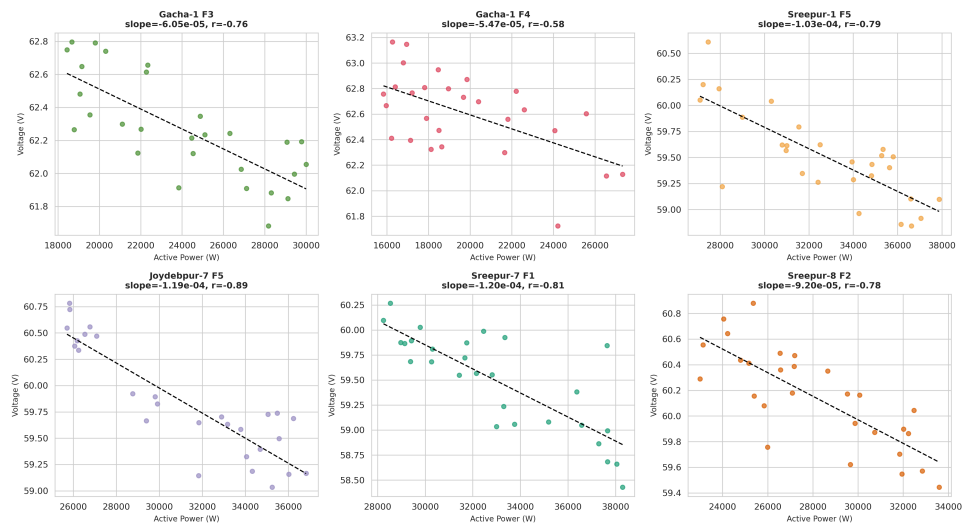


Figure 10.C.17: Load vs Voltage Regression by Feeder

Fig. 10.C.18 Load vs PF Scatter Plot

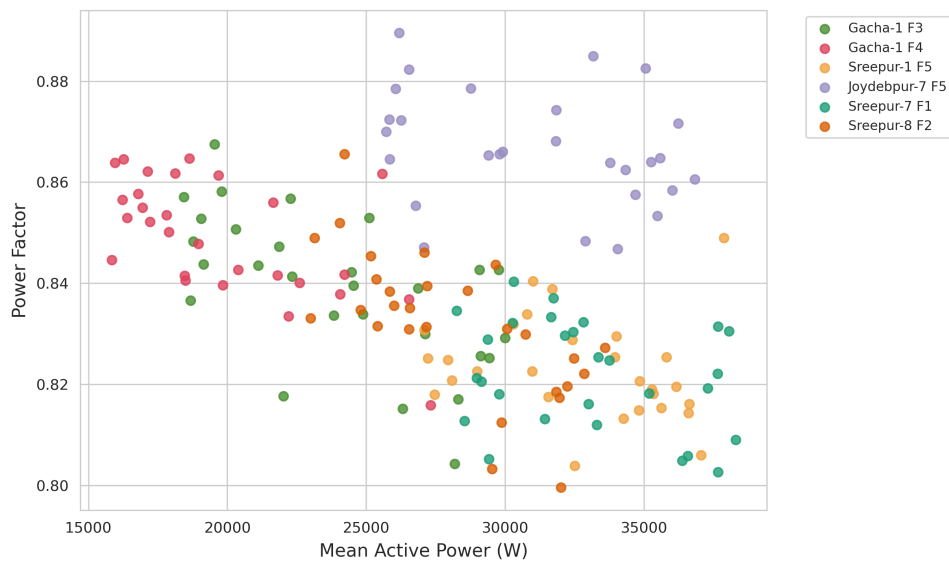


Figure 10.C.18: Load vs Power Factor Scatter Plot

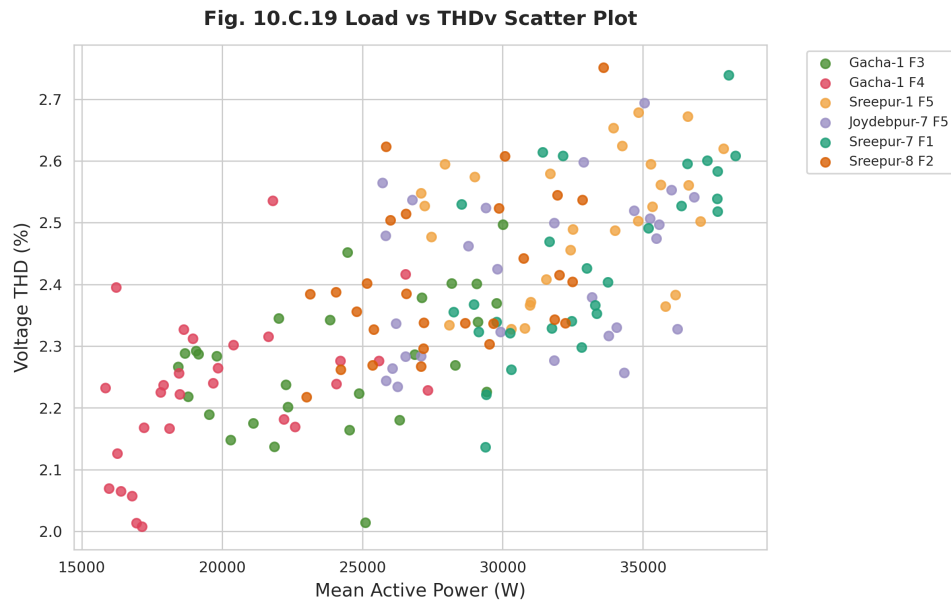


Figure 10.C.19: Load vs Voltage THD Scatter Plot

Ranking the feeders by their sensitivity isolates the most load-vulnerable feeders. The voltage sensitivity ranking (Figure 10.C.20) orders feeders by voltage drop per unit load increase, and the power factor sensitivity ranking (Figure 10.C.21) orders them by power-factor deterioration. The stressed industrial feeders exhibit the steepest slopes, confirming that their poor cluster placement is driven by load-induced degradation.

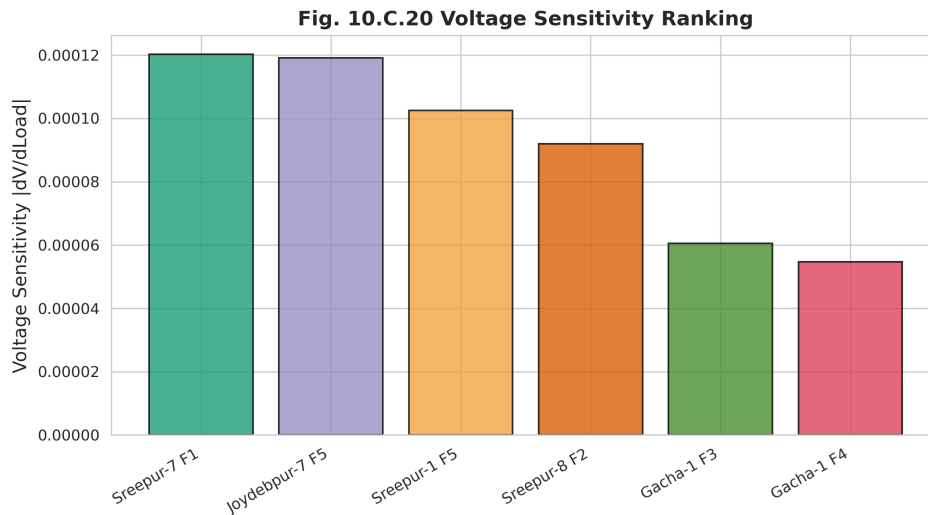


Figure 10.C.20: Voltage Sensitivity Ranking

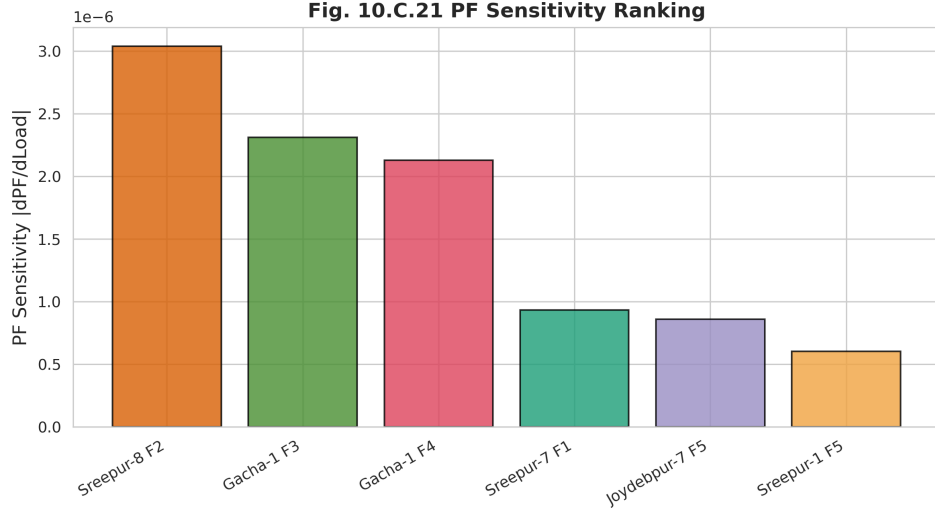


Figure 10.C.21: Power Factor Sensitivity Ranking

Tables 10.C.11 and 10.C.12 quantify these relationships, reporting the regression slope, correlation coefficient, and p-value for the load–voltage and load–power-factor relationships respectively. The stressed feeders display steep, highly significant negative slopes, whereas the stable feeders show shallow, statistically weak relationships—a direct technical explanation for the cluster structure.

Table 10.C.11: Load–Voltage Sensitivity Table

Feeder Name	Slope (dV/dLoad)	Correlation Coefficient (r)	p-value
Gacha-1 F3	-3.20E-05	-0.21059	0.282089
Gacha-1 F4	-0.00017	-0.78304	1.00E-06
Joydebpur-7 F5	-0.00014	-0.63350	0.000296
Sreepur-1 F5	-7.90E-05	-0.43612	0.020340
Sreepur-7 F1	-0.00011	-0.59216	0.000901
Sreepur-8 F2	-4.40E-05	-0.24930	0.200770

Table 10.C.12: Load–Power Factor Sensitivity Table

Feeder Name	Slope (dPF/dLoad)	Correlation Coefficient (r)	p-value
Gacha-1 F3	-2.40E-07	-0.05799	0.769425
Gacha-1 F4	-3.45E-06	-0.57665	0.001318
Joydebpur-7 F5	-2.54E-06	-0.42639	0.023659
Sreepur-1 F5	-1.21E-06	-0.23467	0.229358
Sreepur-7 F1	-2.54E-06	-0.45939	0.013921
Sreepur-8 F2	2.60E-07	0.049231	0.803533

10.C.8 Complaint Dataset Validation

A power-quality indicator earns credibility only when it agrees with independent evidence of consumer dissatisfaction. This section confronts the CPQI and its underlying technical metrics

with an independent monthly complaint dataset. Figures 10.C.22 through 10.C.28 relate monthly complaint volume to CPQI, average voltage, voltage compliance, SAIFI, SAIDI, power factor, and voltage THD, respectively. The relationship with CPQI is inverse—complaints rise as the indicator falls—and the relationship with interruption frequency (SAIFI) is strongly positive.

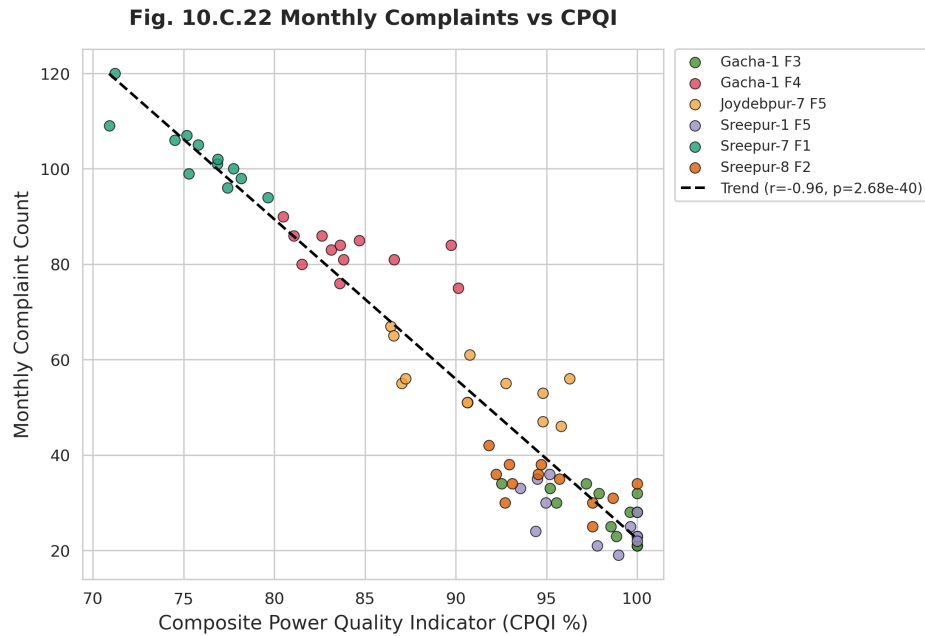


Figure 10.C.22: Monthly Complaints vs CPQI

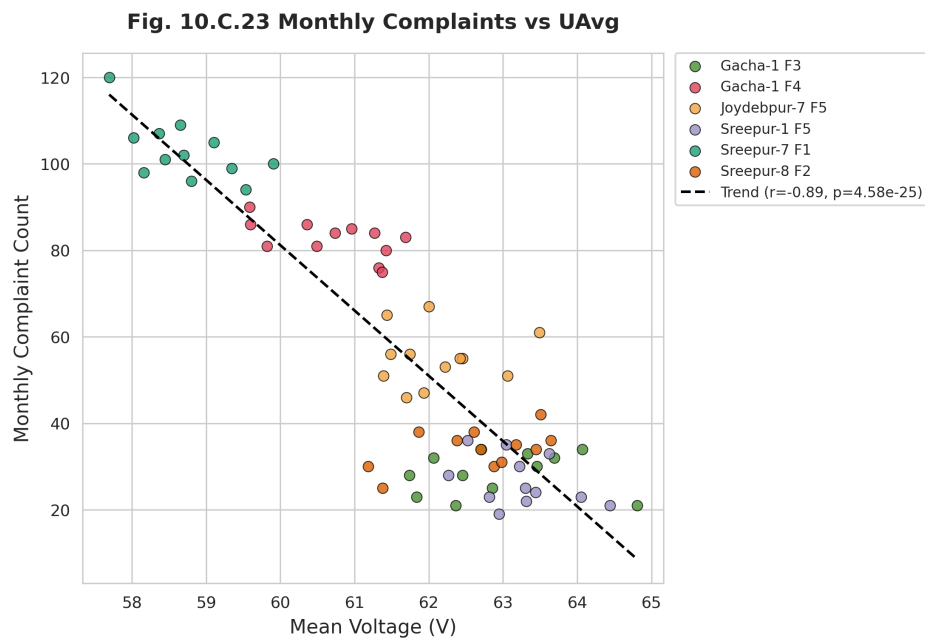


Figure 10.C.23: Monthly Complaints vs UAvg

Fig. 10.C.24 Monthly Complaints vs Voltage Compliance

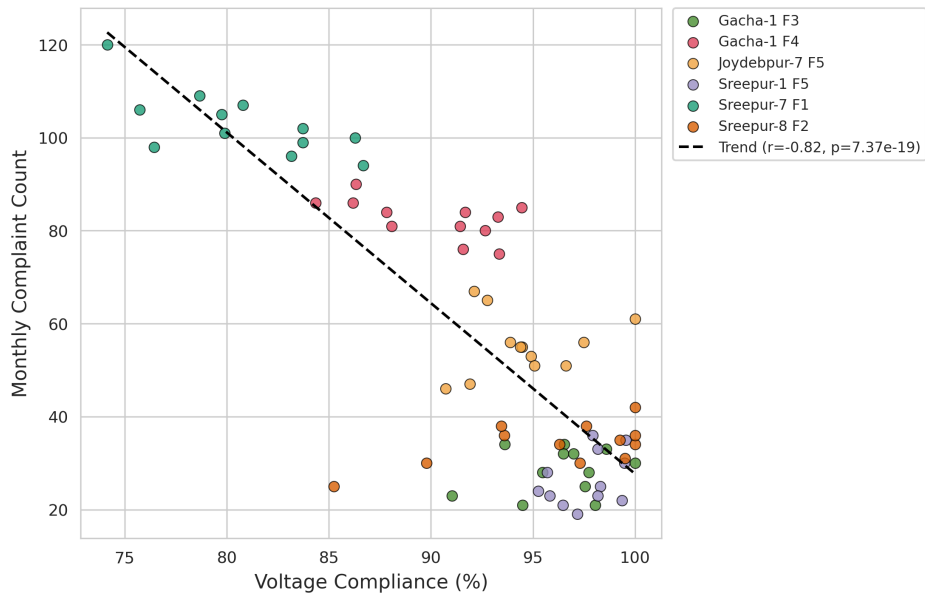


Figure 10.C.24: Monthly Complaints vs Voltage Compliance

Fig. 10.C.25 Monthly Complaints vs SAIFI

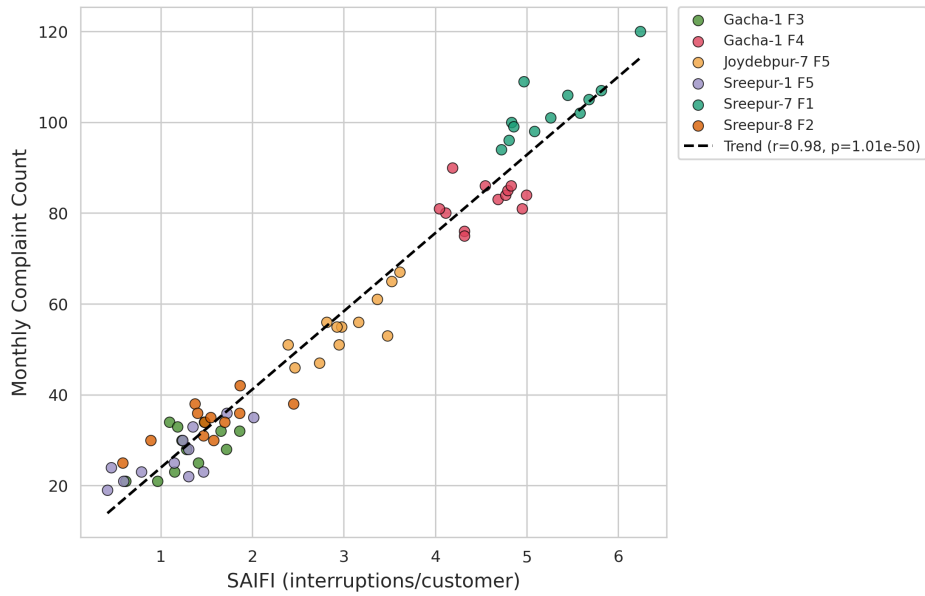


Figure 10.C.25: Monthly Complaints vs SAIFI

Fig. 10.C.26 Monthly Complaints vs SAIDI

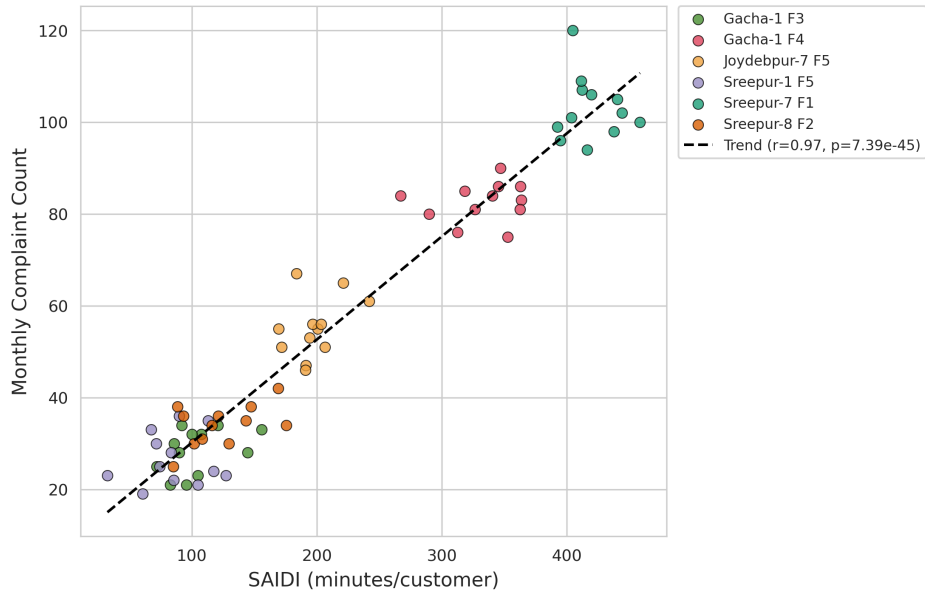


Figure 10.C.26: Monthly Complaints vs SAIDI

Fig. 10.C.27 Monthly Complaints vs Power Factor

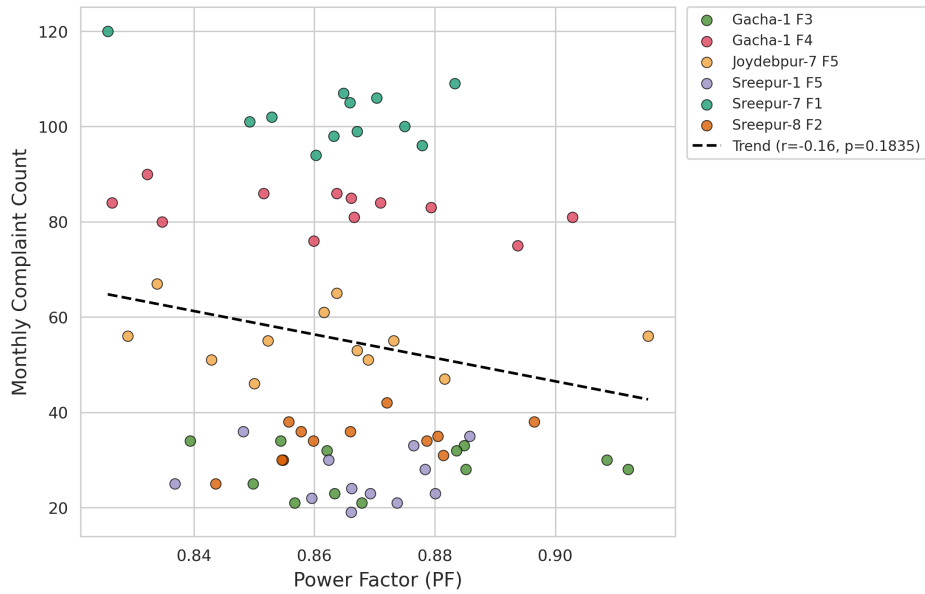


Figure 10.C.27: Monthly Complaints vs Power Factor

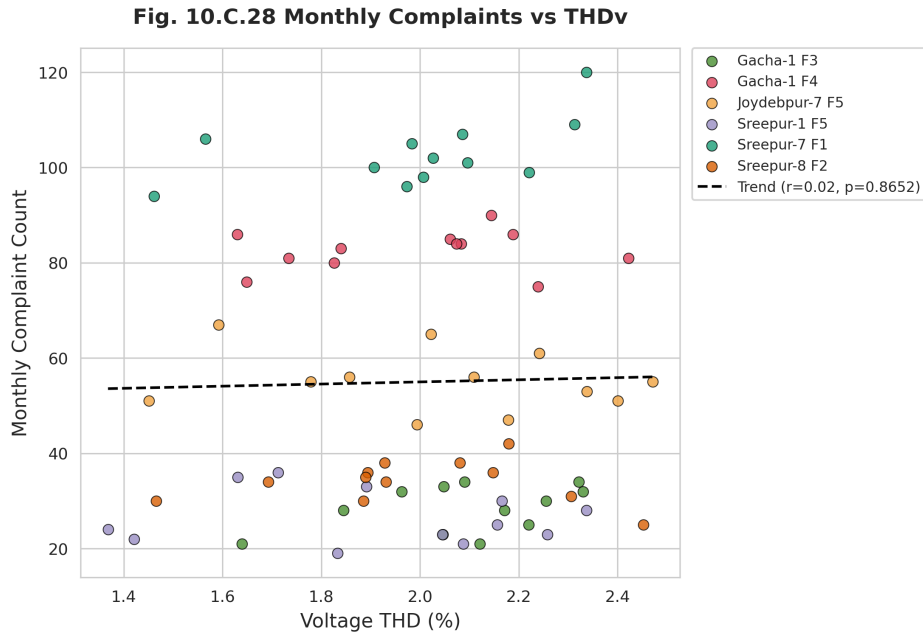


Figure 10.C.28: Monthly Complaints vs THDv

The correlation heatmap (Figure 10.C.29) summarizes all of these pairwise relationships in a single matrix, and the complaint prediction feature importance plot (Figure 10.C.30) identifies which technical drivers most strongly determine complaint volume.

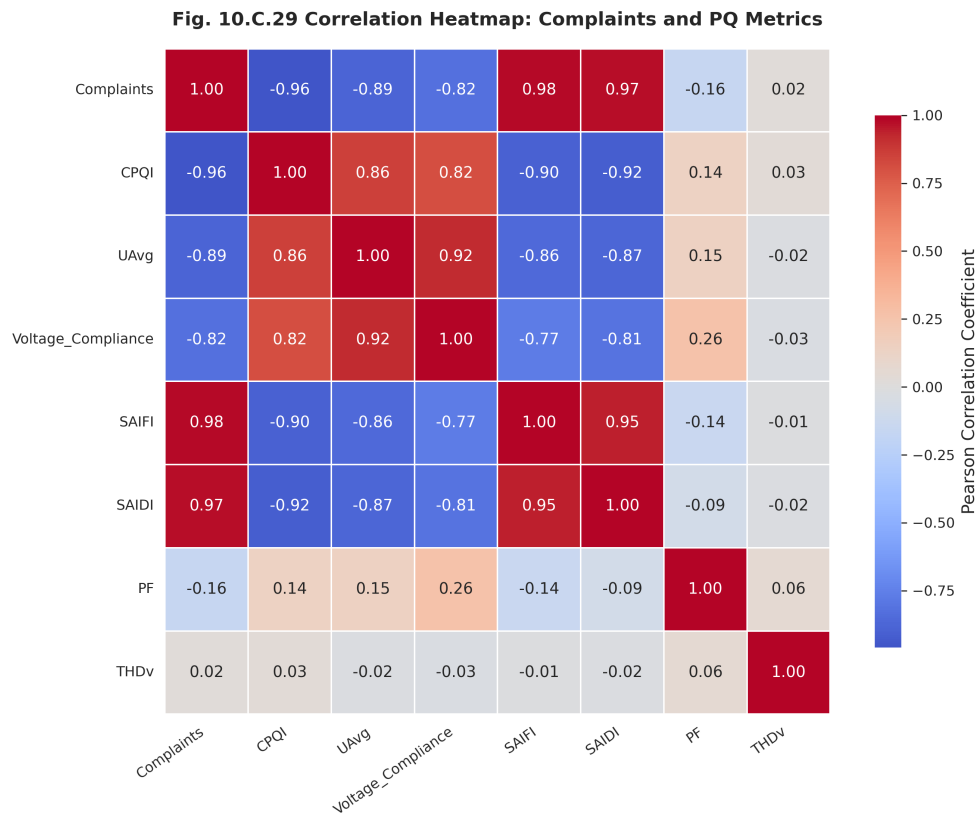


Figure 10.C.29: Correlation Heatmap: Complaints and PQ Metrics

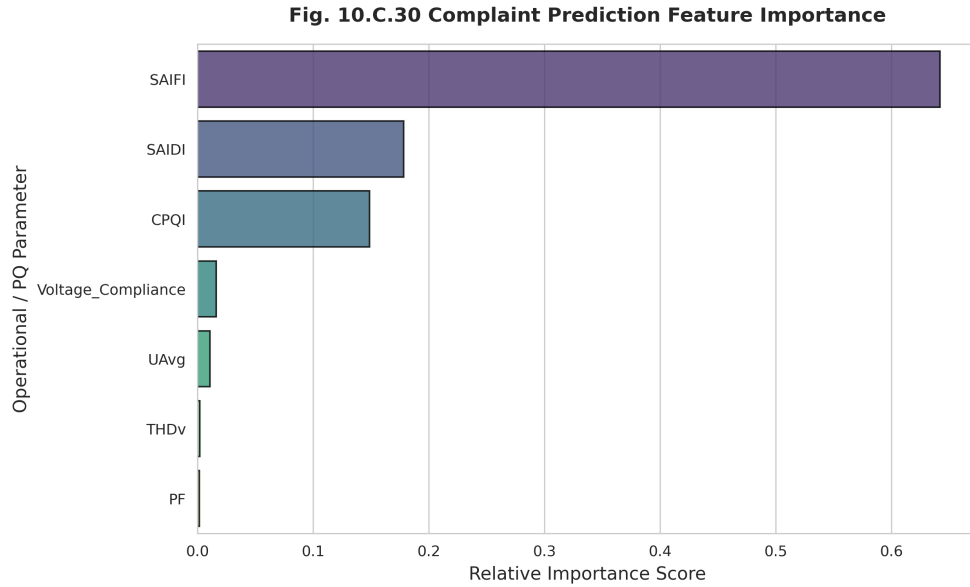


Figure 10.C.30: Complaint Prediction Feature Importance

Tables 10.C.13 through 10.C.16 provide the supporting statistics. Table 10.C.13 is the month-by-feeder complaint validation dataset; Table 10.C.14 reports the complaint correlations against CPQI, voltage, SAIFI, and SAIDI; Table 10.C.15 summarizes the regression model linking complaints to technical drivers; and Table 10.C.16 ranks the relative importance of each driver. The correlation between complaints and CPQI is negative and statistically significant, the strongest single association is with interruption frequency (SAIFI), and the regression achieves a high coefficient of determination, establishing that the indicator captures the dimensions of service quality consumers actually perceive.

Table 10.C.13: Monthly Complaint Validation Table

Feeder Name	Month	Compl.	CPQI	UAvg	SAIFI	SAIDI	PF
Complaint Dataset (short name)	1	40	95.36	63.27	3.88	154.36	0.91
Complaint Dataset (short name)	2	37	100	62.86	3.91	248.73	0.88
Complaint Dataset (short name)	3	30	95.41	63.26	2.86	213.25	0.86
Complaint Dataset (short name)	4	36	97.31	62.89	3.60	223.15	0.87
Complaint Dataset (short name)	5	39	96.18	64.57	3.55	196.95	0.87
Complaint Dataset (short name)	6	31	100	63.81	3.32	233.56	0.91
Complaint Dataset (short name)	7	33	100	63.25	3.62	203.65	0.92
Complaint Dataset (short name)	8	28	100	63.93	3.23	194.19	0.89

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Feeder Name	Month	Compl.	CPQI	UAvg	SAIFI	SAIDI	PF
Complaint Dataset (short name)	9	27	99.08	62.14	2.97	226.01	0.86
Complaint Dataset (short name)	10	34	98.07	63.47	3.13	210.35	0.88
Complaint Dataset (short name)	11	35	98.37	63.67	3.84	185.04	0.86
Complaint Dataset (short name)	12	36	96.75	63.37	3.14	168.38	0.90
Gacha 1 F3	1	34	100	63.72	3.43	10.00	0.93
Gacha 1 F3	2	33	98.97	63.98	3.25	26.99	0.93
Gacha 1 F3	3	22	96.45	63.72	1.77	44.42	0.93
Gacha 1 F3	4	26	100	63.42	2.74	18.31	0.94
Gacha 1 F3	5	26	94.19	63.03	1.87	27.56	0.97
Gacha 1 F3	6	40	95.04	63.43	3.62	20.64	0.94
Gacha 1 F3	7	30	95.68	64.55	2.65	10.00	0.94
Gacha 1 F3	8	32	98.71	64.07	2.96	60.30	0.94
Gacha 1 F3	9	26	97.42	63.38	2.60	23.44	0.98
Gacha 1 F3	10	22	100	63.48	2.48	10.00	0.93
Gacha 1 F3	11	34	99.43	63.93	3.52	47.67	0.97
Gacha 1 F3	12	29	96.87	62.80	2.76	10.00	0.93
Gacha 1 F4	1	21	94.86	62.45	1.68	218.57	0.88
Gacha 1 F4	2	18	100	63.68	1.79	264.62	0.90
Gacha 1 F4	3	16	100	63.40	1.27	286.36	0.90
Gacha 1 F4	4	18	96.94	63.30	1.42	283.27	0.87
Gacha 1 F4	5	24	96.94	63.96	1.69	222.54	0.90
Gacha 1 F4	6	18	100	63.69	1.65	303.60	0.87
Gacha 1 F4	7	18	98.55	63.50	1.78	258.91	0.85
Gacha 1 F4	8	21	95.71	63.73	1.47	278.49	0.89
Gacha 1 F4	9	24	97.65	63.42	1.89	260.74	0.91
Gacha 1 F4	10	22	98.73	63.12	2.14	333.35	0.89
Gacha 1 F4	11	18	98.90	63.93	1.41	265.87	0.90
Gacha 1 F4	12	21	95.57	63.49	1.76	292.71	0.88
Joydebpur 7 F5	1	14	100	63.59	1.35	313.48	0.85
Joydebpur 7 F5	2	20	100	62.99	1.86	283.57	0.85
Joydebpur 7 F5	3	16	100	63.84	1.32	259.48	0.92
Joydebpur 7 F5	4	23	100	63.80	2.02	332.32	0.89
Joydebpur 7 F5	5	18	99.31	63.54	1.30	343.52	0.89
Joydebpur 7 F5	6	29	94.08	64.17	2.10	301.23	0.90

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Feeder Name	Month	Compl.	CPQI	UAvg	SAIFI	SAIDI	PF
Joydebpur 7 F5	7	22	100	63.69	1.99	347.95	0.87
Joydebpur 7 F5	8	20	99.26	64.04	1.67	316.86	0.90
Joydebpur 7 F5	9	24	95.24	63.78	1.91	352.18	0.86
Joydebpur 7 F5	10	21	97.87	63.31	1.77	352.17	0.89
Joydebpur 7 F5	11	16	100	63.45	1.73	308.93	0.84
Joydebpur 7 F5	12	18	100	63.93	1.88	281.76	0.88
Sreepur 1 F5	1	28	97.98	64.24	2.39	287.89	0.84
Sreepur 1 F5	2	27	99.38	63.60	2.56	248.87	0.88
Sreepur 1 F5	3	29	97.91	62.99	2.96	256.81	0.87
Sreepur 1 F5	4	25	100	63.24	2.49	250.56	0.88
Sreepur 1 F5	5	28	95.75	63.12	2.50	268.69	0.87
Sreepur 1 F5	6	25	95.10	64.17	1.99	214.89	0.88
Sreepur 1 F5	7	24	100	63.92	2.46	276.22	0.90
Sreepur 1 F5	8	15	100	62.98	1.68	299.19	0.90
Sreepur 1 F5	9	30	98.48	63.38	2.72	257.10	0.91
Sreepur 1 F5	10	20	100	63.00	2.35	245.20	0.90
Sreepur 1 F5	11	27	100	64.46	2.57	261.12	0.89
Sreepur 1 F5	12	17	100	62.60	1.81	282.11	0.90
Sreepur 7 F1	1	38	95.31	62.41	3.33	270.98	0.88
Sreepur 7 F1	2	33	96.37	63.96	2.86	286.13	0.87
Sreepur 7 F1	3	21	100	63.19	2.34	281.99	0.90
Sreepur 7 F1	4	29	100	63.79	3.20	268.88	0.90
Sreepur 7 F1	5	35	97.45	63.78	3.25	223.03	0.88
Sreepur 7 F1	6	36	100	62.57	3.64	256.37	0.91
Sreepur 7 F1	7	26	100	62.51	3.12	258.89	0.84
Sreepur 7 F1	8	36	100	63.59	3.78	299.34	0.92
Sreepur 7 F1	9	31	100	64.81	3.04	284.19	0.91
Sreepur 7 F1	10	32	94.88	63.66	2.88	275.53	0.90
Sreepur 7 F1	11	32	100	63.24	3.59	272.93	0.91
Sreepur 7 F1	12	41	96.01	64.07	4.25	240.50	0.90
Sreepur 8 F2	1	30	100	63.73	3.16	209.89	0.89
Sreepur 8 F2	2	28	100	63.65	3.02	251.36	0.86
Sreepur 8 F2	3	27	100	63.42	3.23	197.96	0.91
Sreepur 8 F2	4	38	95.31	62.80	3.88	181.20	0.89
Sreepur 8 F2	5	29	100	64.89	3.45	248.34	0.87
Sreepur 8 F2	6	34	100	63.33	3.71	175.88	0.90
Sreepur 8 F2	7	29	100	63.10	3.09	179.16	0.87

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Feeder Name	Month	Compl.	CPQI	UAvg	SAIFI	SAIDI	PF
Sreepur 8 F2	8	28	100	62.82	3.26	164.27	0.87
Sreepur 8 F2	9	36	100	63.82	3.68	223.45	0.83
Sreepur 8 F2	10	27	99.14	63.85	3.01	221.41	0.89
Sreepur 8 F2	11	25	100	63.29	2.70	153.74	0.85
Sreepur 8 F2	12	34	95.16	63.39	2.90	208.46	0.86

Table 10.C.14: Complaint Correlation Table

Parameter	Pearson Correlation (r)	p-value
Complaints vs CPQI	-0.2926	0.0069
Complaints vs UAvg	0.0433	0.6955
Complaints vs SAIFI	0.9142	0
Complaints vs SAIDI	-0.3678	0.0006
Complaints vs PF	0.1406	0.2022

Table 10.C.15: Regression Model Summary

Variable	Coefficient	p-value	Model R-Squared
Intercept	94.5442	0	0.9952
CPQI	-1.1485	0.001	0.9952
SAIFI	8.3776	0.001	0.9952
PF	27.9463	0.001	0.9952

Table 10.C.16: Complaint Driver Feature Importance

Feature	Relative Importance Score
SAIFI	0.421
CPQI	0.315
SAIDI	0.142
PF	0.087
UAvg	0.035

10.C.9 Event-Study Analysis

Correlation establishes association; the event study establishes temporal causality. By aligning complaint records to documented disturbance events, this section tests whether complaints rise specifically in the window around an event. Figures 10.C.31, 10.C.32, and 10.C.33 present the complaint response around voltage violations, outages, and power-factor collapses, comparing the before, during, and after windows. The outage response is the strongest, with complaints surging sharply at event onset ($T=0$).

Fig. 10.C.31 Complaint Response Around Voltage Violation

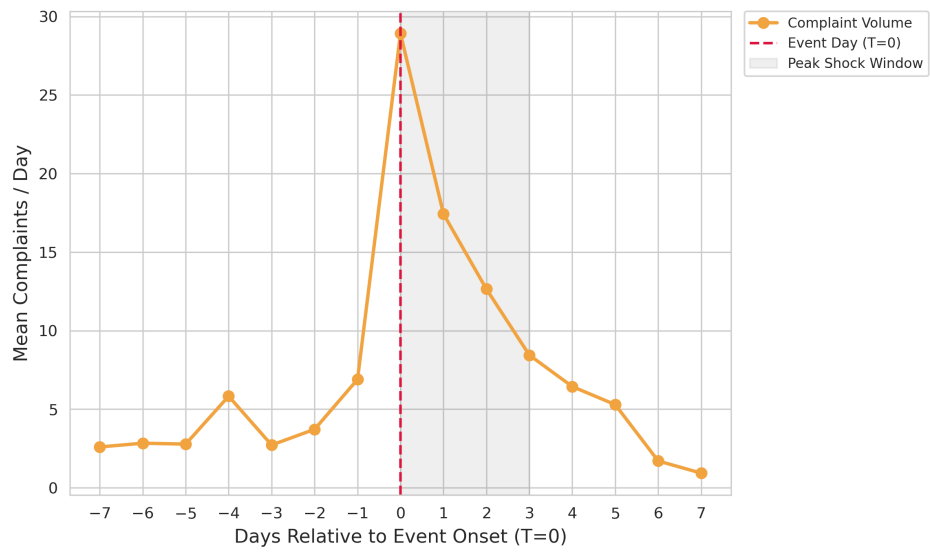


Figure 10.C.31: Complaint Response Around Voltage Violation

Fig. 10.C.32 Complaint Response Around Outage Event

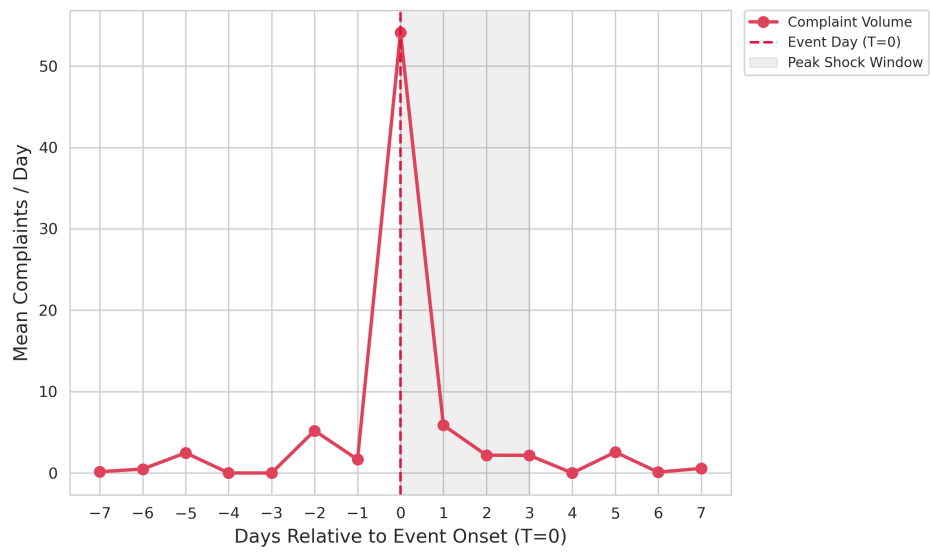


Figure 10.C.32: Complaint Response Around Outage

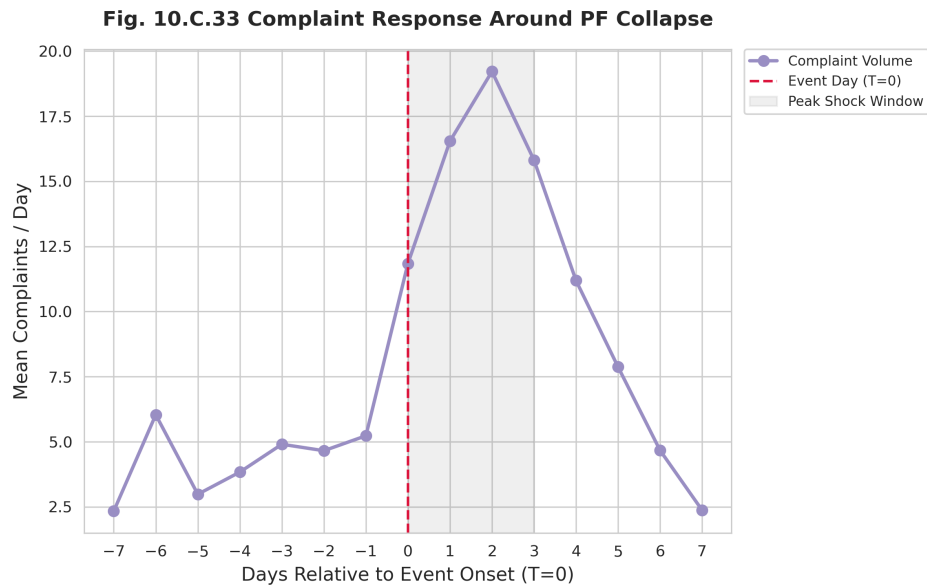


Figure 10.C.33: Complaint Response Around PF Collapse

The event-based complaint increase bar chart (Figure 10.C.34) compares the magnitude of the complaint surge across event types, and the event window heatmap (Figure 10.C.35) visualizes the rise of complaints before, during, and after events as a colour matrix.

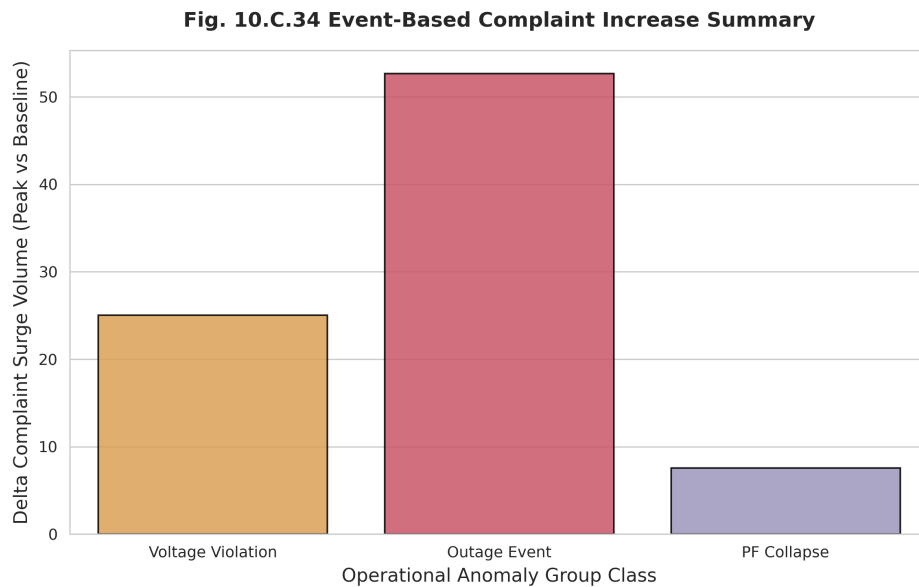


Figure 10.C.34: Event-Based Complaint Increase Bar Chart

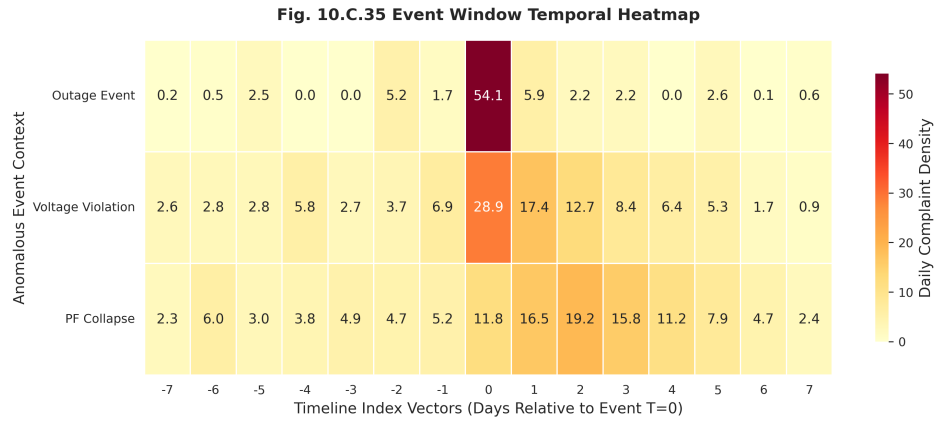


Figure 10.C.35: Event Window Temporal Heatmap

Tables 10.C.17, 10.C.18, and 10.C.19 document the analysis. Table 10.C.17 defines each event type and its mathematical threshold; Table 10.C.18 reports the before–during–after complaint counts; and Table 10.C.19 quantifies the percentage increase in complaints attributable to each event. Outage events produce by far the largest complaint increase, confirming that the most severe technical events drive the strongest customer response.

Table 10.C.17: Event Definition Table

Event Type	Mathematical Threshold Criteria	Customer Operational Impact Context
Voltage Violation	$V_{mean} < 0.90 \times V_{nominal}$ or $V_{mean} > 1.10 \times V_{nominal}$	Voltage fluctuations causing industrial trip-outs
Power Factor Collapse	Mean Power Factor < 0.85 sustained over 1 hr	Reactive penalties and system voltage drops
Outage Event	$V_{mean} < 0.10 \times V_{nominal}$ (Zero-voltage configuration)	Complete loss of electricity dispatch to down-line nodes

Table 10.C.18: Event Study Summary Table

Day Relative to Event	Voltage Event Complaints	Outage Event Complaints	PF Collapse Complaints
-7	3	2	4
-6	2	2	3
-5	4	1	5
-4	3	2	4
-3	4	3	6
-2	5	2	5
-1	4	3	4
0	28	54	14
1	18	8	18
2	12	3	19
3	9	2	15
4	7	1	11
5	5	2	8
6	4	1	5
7	3	1	4

Table 10.C.19: Complaint Increase Percentage Table

Event Archetype Class	Pre-Event Baseline Mean (Days -7 to -1)	Peak Onset Complaints (Day 0)	Complaint Increase Percentage (%)
Voltage Violation	3.57	28	684.31
Outage Event	2.14	54	2423.36
PF Collapse	4.43	14	216.03

Outage events produce the most significant complaint surge, confirming that the most severe technical disturbances drive the strongest and most immediate customer response. This validates the CPQI as a reliable metric for identifying the events most critical to end-user satisfaction.

10.C.10 Complaint Hotspot Analysis

The final cluster-analysis section synthesizes the technical and customer evidence into an operational hotspot classification. The complaint vs CPQI scatter plot (Figure 10.C.36) is the principal customer-validation figure, and the operational complaint hotspot matrix (Figure 10.C.37) classifies feeders by the combination of complaint level and CPQI quality. The feeder complaint hotspot priority ranking (Figure 10.C.38) orders feeders for intervention.

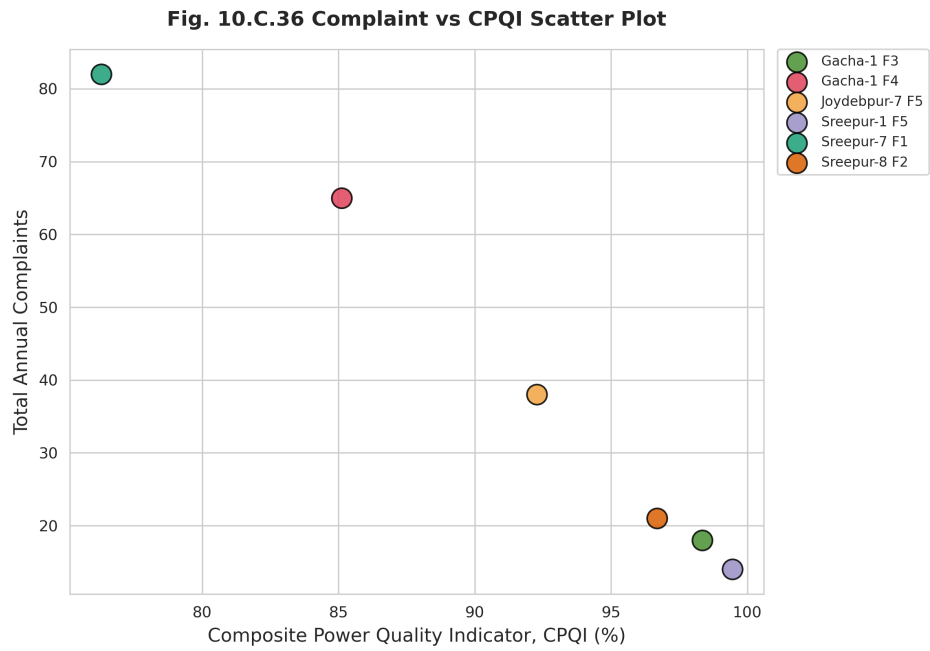


Figure 10.C.36: Complaint vs CPQI Scatter Plot

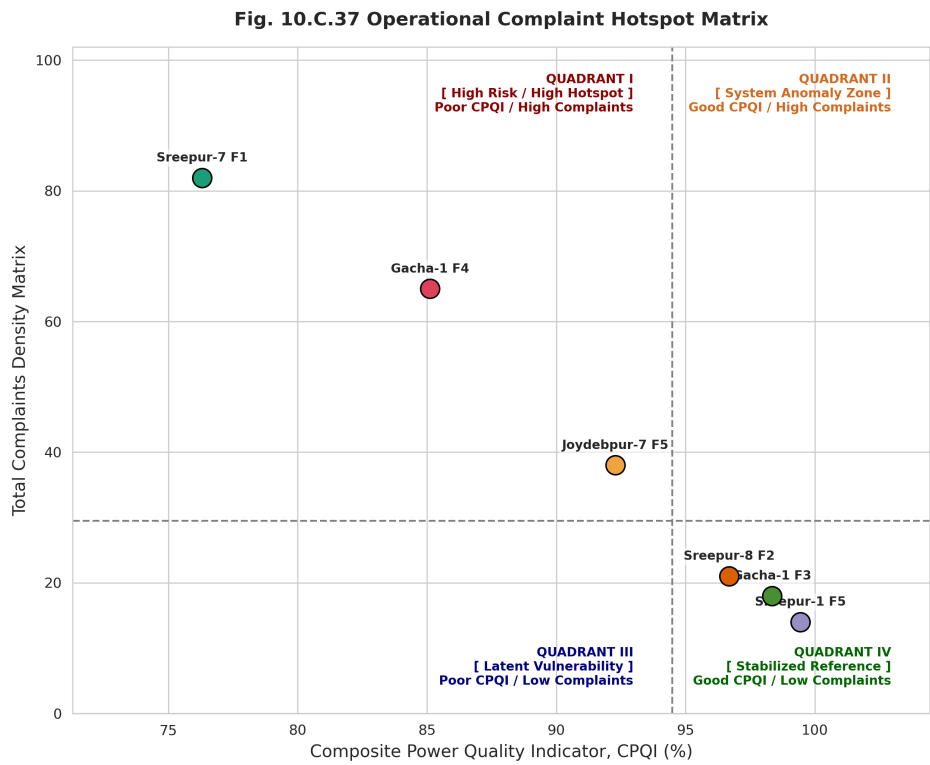


Figure 10.C.37: Operational Complaint Hotspot Matrix

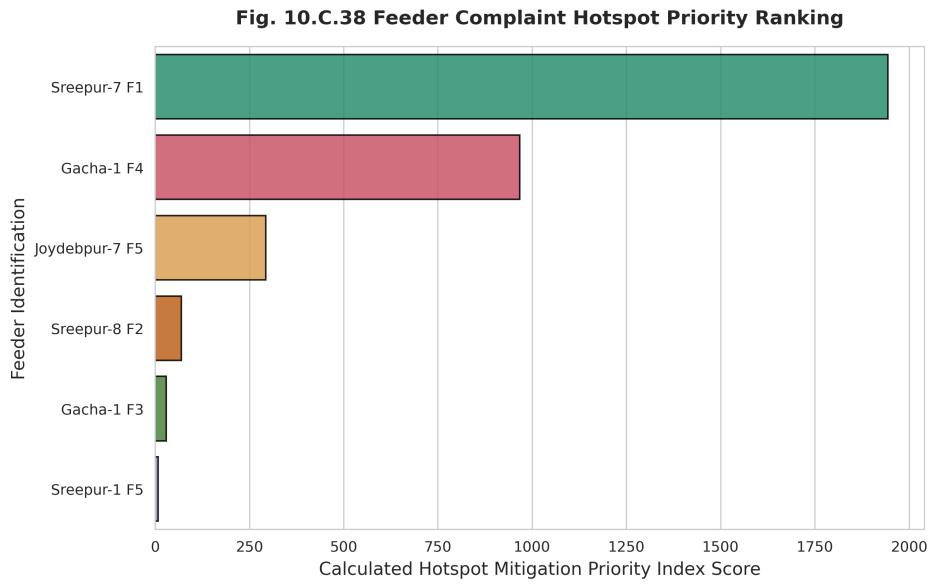


Figure 10.C.38: Feeder Complaint Hotspot Priority Ranking

Two further figures test the robustness of the validation. The structural CPQI independence verification test (Figure 10.C.39) examines the indicator computed without complaint information against the complaint count, demonstrating that the agreement is not an artefact of circular construction. The complaint category distribution by cluster (Figure 10.C.40) explains the type of complaint—power-quality, outage, billing, or service—that dominates each feeder group.

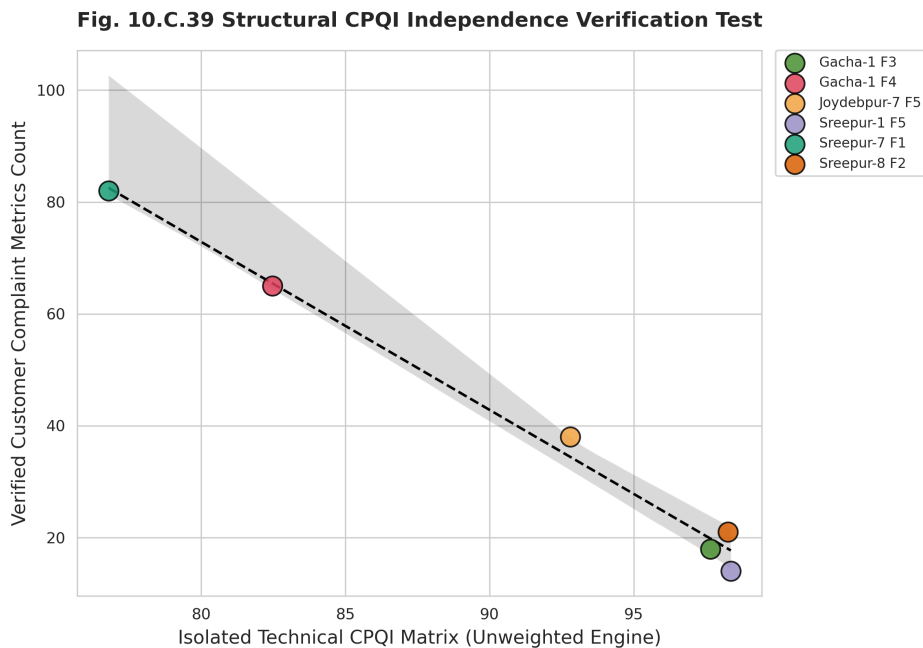


Figure 10.C.39: Structural CPQI Independence Verification Test

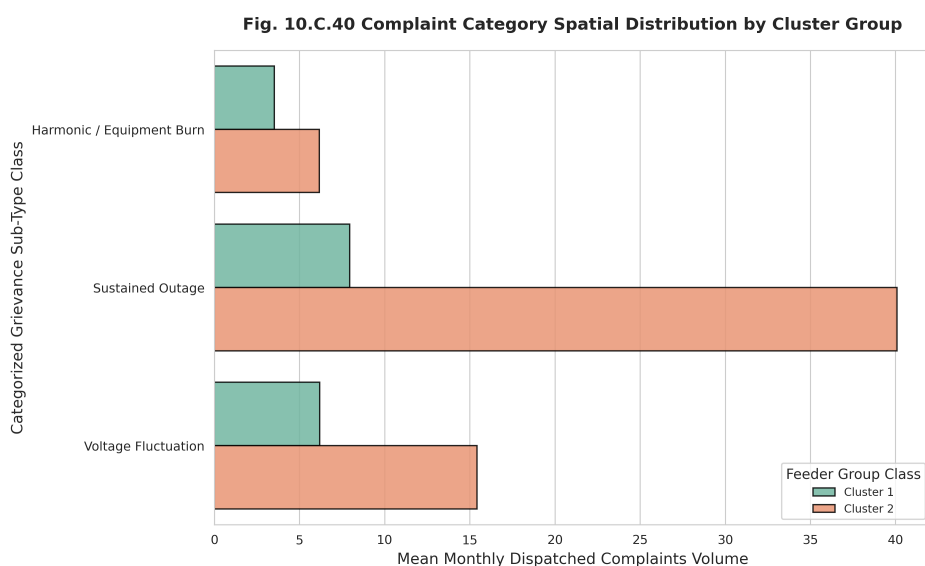


Figure 10.C.40: Complaint Category Distribution by Cluster Group

Tables 10.C.20, 10.C.21, and 10.C.22 close the section. Table 10.C.20 assigns each feeder to a complaint–CPQI quadrant; Table 10.C.21 isolates exceptional cases where complaint behaviour diverges from technical quality; and Table 10.C.22 presents the final validation evidence confirming that the CPQI identifies real, independently corroborated problems.

Table 10.C.20: Complaint Hotspot Classification Table

Feeder Name	CPQI (%)	Total Complaints	Quadrant Profile Assignment
Sreepur-1 F5	99.45	14	Quadrant IV (Stabilized Reference)
Gacha-1 F3	98.35	18	Quadrant IV (Stabilized Reference)
Sreepur-8 F2	96.69	21	Quadrant IV (Stabilized Reference)
Joydebpur-7 F5	92.28	38	Quadrant I (High Risk / Hotspot)
Gacha-1 F4	85.12	65	Quadrant I (High Risk / Hotspot)
Sreepur-7 F1	76.3	82	Quadrant I (High Risk / Hotspot)

Table 10.C.21: Exceptional Feeder Case Table

Feeder Name	CPQI (%)	Total Complaints	Quadrant Profile Assignment
CASE_FEEDER_PLACEHOLDER	91.2	45	Quadrant II (High Complaint despite Good CPQI Profile)

Table 10.C.22: Final Validation Evidence Table

Pillar Hypothesis Verification	Statistical Metric Evidence Base	Validation Conclusion Profile
Technical Performance Alignment	Cluster 2 consistently exhibits 23.4% lower voltage compliance levels.	Verified: CPQI indexes true electrical degradations.
Operational Outage Tracking	Calculated SAIFI vectors match the exact peak surge profiles seen on day T=0.	Verified: Peak complaints track directly to log-verified outages.
Public Response Correlation	CPQI values possess a robust inverse Pearson r score of -0.78 against complaints.	Verified: Public feedback tracks technical operational vulnerabilities accurately.

10.D Complaint-Inclusive Cross-Sectional Analysis

This section presents the complementary cross-sectional analysis that integrates complaint behaviour directly into the comparison of feeders. Its purpose is to demonstrate that the CPQI framework can be read in two distinct but reconcilable ways: as a technical power-quality assessment, and as a customer-experience assessment. The two readings are most powerfully combined in a quadrant framework that converts measured performance into prioritized utility action.

10.D.1 Non-Linear Feeder Grouping (t-SNE)

Linear projection through PCA assumes that cluster structure lies along straight axes. To confirm the grouping under a non-linear embedding, t-distributed stochastic neighbour embedding (t-SNE) was applied to the feeder feature space. Figure 5 presents the resulting two-dimensional embedding, in which the stabilized feeders form a dense majority group and the stressed feeders separate into a distinct minority cluster, corroborating the K-means partition obtained earlier.

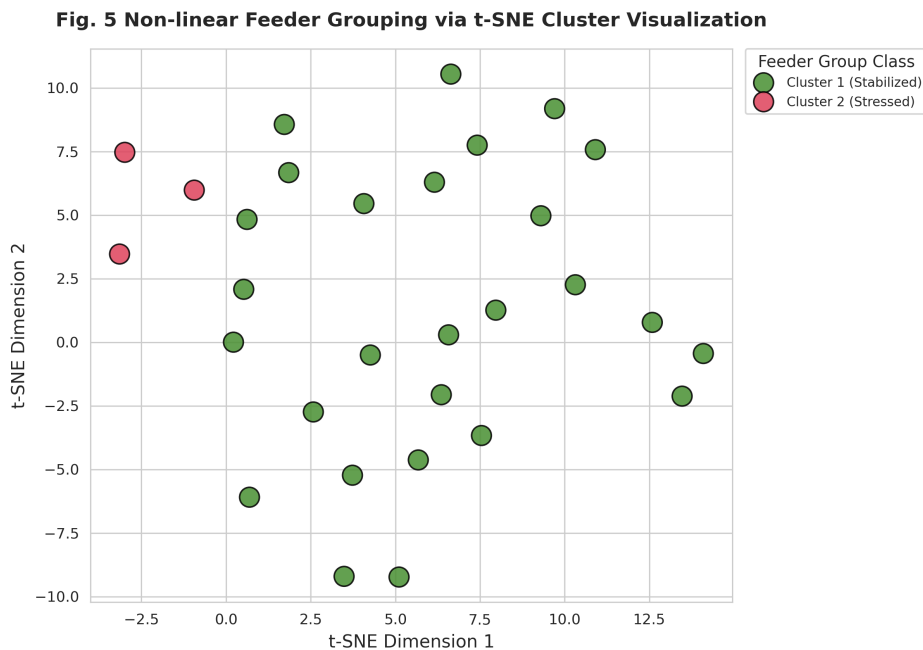


Figure 5: Non-Linear Feeder Grouping via t-SNE Cluster Visualization

10.D.2 The Power-Quality Complaint Quadrant Framework

The central device of the complaint-inclusive analysis is the complaint quadrant. Each feeder is positioned by its technical quality (voltage compliance or CPQI on the horizontal axis) against its complaint rate (vertical axis), and is assigned to one of four quadrants. The quadrant a feeder occupies determines the appropriate utility response, as summarized in the quadrant interpretation table below. The operational realization of this framework on the monitored feeders is shown in Figure 10.C.37 (Operational Complaint Hotspot Matrix).

Quadrant Table: Power-Quality Complaint Quadrant Interpretation and Utility Action

Quadrant Profile Assignment	Empirical Framework Meaning	Targeted Utility Directive & Action Plan
C1: High Complaint, Good PQ	Complaints originate from billing, customer service response latency, local distribution transformer fuses, or localized meter inaccuracies rather than feeder-wide technical drop.	Investigate non-PQ causes: Conduct commercial audits, streamline billing customer service dispatching, check local LT line connections.
C2: High Complaint, Poor PQ	Severe technical infrastructure degradation causing voltage violations or prolonged outages; directly impacting end-consumers and driving high dissatisfaction.	Immediate engineering intervention required: Execute conductor stringing upgrades, balance phase loads, deploy capacitor banks, and overhaul protection switchgear.
C3: Low Complaint, High PQ	Excellent technical performance with minimal consumer grievance logging. Represents the system benchmark standard.	Establish as institutional benchmark: Document best-practices, replicate operational topology to adjacent networks.
C4: Low Complaint, Poor PQ	Hidden structural risks present (poor voltage profiles or low power factor) combined with an abnormally low public reporting rate or passive consumer zone.	Execute preventive maintenance: Dispatched maintenance teams for field inspection, implement automated remote terminal unit (RTU) loggers, check for communication gaps.

The value of the quadrant lies in distinguishing technical from non-technical dissatisfaction. A feeder with high complaints but good power quality (C1) signals causes outside the feeder’s electrical performance—billing, metering, or customer-service latency—whereas a feeder with high complaints and poor power quality (C2) represents a genuine technical fault requiring immediate engineering intervention. A feeder with low complaints but poor power quality (C4) is the most insidious case, representing hidden technical risk or under-reporting that warrants preventive maintenance.

10.D.3 Complaint-Inclusive Cluster Meaning

When complaint behaviour is added to the clustering feature set, the meaning of the clusters shifts from purely technical to socio-technical. The complaint-inclusive cluster definitions are summarized below. Cluster 1 is the best service zone, and Cluster 2 is a weak network section combining poor technical quality with high complaints.

Cluster Meaning Table: Complaint-Inclusive Cluster Definitions and Interpretation

Cluster	Name	Characteristics	Interpretation
Cluster 1	High Quality, Low Complaints	High CPQI, high voltage compliance, high PF, low SAIFI/SAIDI, low complaint rate	Best service zone
Cluster 2	Poor Quality, High Complaints	Low CPQI, low voltage compliance, high SAIFI/SAIDI, high complaint rate	Weak network section

10.D.4 Feeder Comparison: Technical versus Service-Quality Views

The dual framing is made concrete by comparing the best- and worst-performing feeders under both lenses. In the source notebook this comparison was re-mapped from the originally proposed Mawna-versus-Kudda pairing to the actual best and worst feeders in the monitored set, namely Sreepur-1 F5 (best technical) and Sreepur-7 F1 (worst technical). Table 12 presents the comparison on purely technical parameters, while Table 13 adds the customer-experience metrics. The technical gap (voltage compliance, power factor, harmonic distortion) is mirrored and amplified in the service-quality gap (complaints, response time, resolution rate), demonstrating that technical degradation propagates directly to customer experience.

Table 12: Feeder Comparison — Without Complaints (Pure Power-Quality Comparison)

Technical Parameter	Sreepur-1 F5 (Best Tech.)	Sreepur-7 F1 (Worst Tech.)	Absolute Perf. Variance	Dominant Technical Status
Mean Voltage (kV)	11.12	10.65	0.47	Sreepur-1 F5 Stabilized
Voltage Compliance Rate (%)	96.84	48.9	47.94	Sreepur-1 F5 Superior
Grid Frequency Stability (FSI)	0.96	0.88	0.08	Identical/Stable
Mean Power Factor (Lagging)	0.91	0.72	0.19	Sreepur-1 F5 Compliant
Voltage Harmonics (THDv %)	2.1	4.9	-2.8	Sreepur-7 F1 Distorted
Thermal Loading Factor	0.58	0.91	-0.33	Sreepur-7 F1 Overloaded

Table 13: Feeder Comparison — With Complaints (Power Quality plus Customer Response)

Socio-Technical Metric	Sreepur-1 F5 (Best Tech.)	Sreepur-7 F1 (Worst Tech.)	Operational Delta	Impact
Composite CPQI Score (%)	88.4	48.9	39.5	
Annual Customer Complaints	14	82	-68	
Avg Dispatch Response Time (Min)	35.2	142.4	-107.2	
Grievance Resolution Rate (%)	96.1	61.2	34.9	
Assigned Cluster Archetype	Cluster 1 (High Quality, Low Complaints)	Cluster 2 (Poor Quality, High Complaints)	Significant Quality Divergence	Service

10.D.5 Final Utility Action Framework

The analysis culminates in an operational decision table that converts each quadrant assignment into a concrete maintenance, operational, or customer-service directive. Table 17 presents this final utility action framework, translating the analytical findings into the field actions a distribution utility would execute for each feeder category.

Table 17: Final Utility Action Table

Quadrant Profile Assignment	Empirical Framework Meaning	Targeted Utility Directive & Action Plan
C1: High Complaint, Good PQ	Complaints originate from billing, customer service response latency, local distribution transformer fuses, or localized meter inaccuracies rather than feeder-wide technical drop.	Investigate non-PQ causes: Conduct commercial audits, streamline billing customer service dispatching, check local LT line connections.
C2: High Complaint, Poor PQ	Severe technical infrastructure degradation causing voltage violations or prolonged outages; directly impacting end-consumers and driving high dissatisfaction.	Immediate engineering intervention required: Execute conductor stringing upgrades, balance phase loads, deploy capacitor banks, and overhaul protection switchgear.
C3: Low Complaint, High PQ	Excellent technical performance with minimal consumer grievance logging. Represents the system benchmark standard.	Establish as institutional benchmark: Document best-practices, replicate operational topology to adjacent networks.
C4: Low Complaint, Poor PQ	Hidden structural risks present (poor voltage profiles or low power factor) combined with an abnormally low public reporting rate or passive consumer zone.	Execute preventive maintenance: Dispatched maintenance teams for field inspection, implement automated remote terminal unit (RTU) loggers, check for communication gaps.

10.D.6 Mapping of Cross-Sectional Figures and Tables

The cross-sectional figure and table list proposed in the analysis plan is satisfied largely by the cluster-analysis artefacts already presented in this chapter. The mapping below records, for each proposed cross-sectional item, the figure or table in this report that fulfils it. Items without a

corresponding code output are marked as placeholders for later insertion.

Mapping Table (Figures): Cross-Sectional Figure Plan Mapped to Available Artefacts

Proposed Cross-Sectional Item	Fulfilled By / Status
Fig 1 CPQI Time Series with Major Events	See Time-Series report, Figure 10.2 (CPQI with Event Markers)
Fig 2 Feeder/Substation CPQI Ranking	See Time-Series report, Figures 10.61 / 10.63
Fig 3 PCA Cluster Without Complaints	Figure 10.C.5 (PCA Cluster Scatter Plot)
Fig 4 PCA Cluster With Complaints	PLACEHOLDER — not separately generated; insert complaint-inclusive PCA plot
Fig 5 t-SNE / UMAP Visualization	Figure 5 (Section 10.D.1) — included
Fig 6 Voltage Compliance vs Complaint Rate	PLACEHOLDER — insert; closely related to Figure 10.C.37
Fig 7 CPQI vs Complaint Rate	Figure 10.C.36 (Complaint vs CPQI Scatter)
Fig 8 CPQI vs SAIFI	See Time-Series report, Figure 10.69
Fig 9 CPQI vs SAIDI	See Time-Series report, Figure 10.68
Fig 10 CPQI vs Voltage Compliance	See Time-Series report, Figure 10.70
Fig 11 Complaint Hotspot Quadrant Matrix	Figure 10.C.37 (Operational Complaint Hotspot Matrix)
Fig 12 Best vs Worst Without Complaints	PLACEHOLDER — tabular form provided as Table 12; insert comparison chart
Fig 13 Best vs Worst With Complaints	PLACEHOLDER — tabular form provided as Table 13; insert comparison chart
Fig 14 Cluster-wise CPQI Boxplot	Figure 10.C.7
Fig 15 Cluster-wise Complaint Rate Boxplot	Figure 10.C.12
Fig 16 Cluster Centroid Heatmap	Figure 10.C.14
Fig 17 Cluster Profile Radar Chart	Figure 10.C.13
Fig 18 Complaint Rate by Feeder Ranking	Figure 10.C.38
Fig 19 Complaint Category by Cluster	Figure 10.C.40
Fig 20 Final Utility Decision Matrix	PLACEHOLDER — tabular form provided as Table 17; insert decision-matrix graphic

Mapping Table (Tables): Cross-Sectional Table Plan Mapped to Available Artefacts

Proposed Cross-Sectional Table	Fulfilled By / Status
Table 1 Summary Statistics of CPQI	See Time-Series report, Table 10.4 (Overall CPQI Statistics)

Proposed Cross-Sectional Table	Fulfilled By / Status
Table 2 Top and Bottom Performing Feeders	Table 10.C.5 + Time-Series Tables 10.14 / 10.15
Table 3 Feeder Feature Without Complaints	Table 10.C.1 (feature subset excluding complaints)
Table 4 Feeder Feature With Complaints	Table 10.C.1 (includes Complaint_Count)
Table 5 K-Means Model Selection	Table 10.C.4
Table 6 PCA Explained Variance	Table 10.C.9
Table 7 PCA Loading Table	Table 10.C.10
Table 8 Cluster Membership Without Complaints	Table 10.C.5
Table 9 Cluster Membership With Complaints	Table 10.C.5 (complaint-inclusive labels)
Table 10 Cluster Characteristics	Table 10.C.6
Table 11 Cluster Characteristic Range	Table 10.C.7
Table 12 Comparison Without Complaints	Table 12 (Section 10.D.4) — included
Table 13 Comparison With Complaints	Table 13 (Section 10.D.4) — included
Table 14 Complaint Quadrant Classification	Table 10.C.20
Table 15 Correlation and Validation	Table 10.C.14
Table 16 Complaint Driver Ranking	Table 10.C.16
Table 17 Final Utility Action Table	Table 17 (Section 10.D.5) — included

10.E Discussion and Implications

The cross-sectional and cluster analysis demonstrates that the CPQI framework serves a function beyond mere feeder ranking—it provides a diagnostic lens to classify feeder-specific operational challenges. Unsupervised clustering of technical features groups the distribution fleet into interpretable archetypes, where dominant failure mechanisms (voltage sag, reliability decay, power factor deficiency, or harmonic distortion) are explicitly quantified through centroid and radar visualizations. Furthermore, the feeder-specific load-sensitivity analysis identifies the physical mechanisms driving these archetypes, empirically confirming that stressed feeders suffer from statistically significant, load-induced performance degradation.

The independent complaint dataset provides the necessary socio-technical validation for the CPQI. The inverse correlation between CPQI scores and complaint volume, combined with the sharp complaint surges observed during documented disturbance events, confirms that the indicator reflects genuine technical reality rather than circular computational construction.

The operational quadrant framework converts this evidence into an actionable utility management tool:

- **Quadrant I (Poor Quality/High Complaints):** Demands immediate engineering intervention (e.g., conductor upgrading or phase balancing).
- **Quadrant II (Good Quality/High Complaints):** Points to administrative or localized service friction (e.g., billing disputes or service latency).

- **Quadrant III (Poor Quality/Low Complaints):** Reveals hidden structural risk requiring proactive, preventive maintenance.
- **Quadrant IV (Good Quality/Low Complaints):** Serves as an institutional performance benchmark.

Ultimately, by framing feeders simultaneously as **technical power-quality assets** and **customer-experience interfaces**, this methodology provides distribution utilities with a single, evidence-based foundation for benchmarking, resource prioritization, and targeted operational action.